The Real Cost of Benchmarking*

Sebastian Hanson
Stanford GSB

Christian Kontz
Stanford GSB

July 2, 2024

Abstract

This paper documents a novel channel through which the benchmarking of fund managers affects firm behavior. Using a difference-in-differences design, we show that increases in a stock's benchmarking intensity cause its CAPM β to rise. Stocks with a benchmarking intensity increase of more than 5 percentage points due to index inclusion subsequently have 15% higher CAPM βs. These changes in CAPM βs are not driven by changes in firm fundamentals and persist for at least seven years. Using an instrumental variable design, we find that firms with higher CAPM βs due to index inclusion reduce investment and have 7.1% less physical capital after six years. Instead of investing, these firms hold more cash and gradually increase payouts to shareholders. Consistent with this mechanism, we show that increasing benchmarking intensity predicts a higher perceived cost of capital by firm managers. From 2006 to 2016, increases in CAPM βs due to higher benchmarking caused a 7.6% reduction in capital accumulation at the aggregate level. Our findings highlight the significant influence of fund manager benchmarking on firm behavior and real quantities through changes in the perceived cost of capital.

JEL classification: D2, E22, G11, G23, G31
Keywords: Benchmarking, Beta, Capital Allocation, CAPM, Cost of Capital, Index Inclusion, Investment, Investment Shortfall, Misallocation, Missing Investment

*We thank Juliane Begenau, Ben Hébert, Arvind Krishnamurthy, Hanno Lustig, Adrien Matray, Amir Sufi, and seminar and conference participants for helpful comments. Christian Kontz thanks the Becker Friedman Institute at the University of Chicago for their hospitality during part of the research process for this paper. All errors remain our own. Corresponding author's email: ckontz@stanford.edu. First version: April 07, 2024

1 Introduction

Recent decades have seen a large increase in capital linked to benchmark stock indices like the S&P 500 and Russell indices. This includes not only passive index funds that track indices by design, but also actively managed funds whose performance is evaluated against a benchmark index. As a result, an increasing share of capital is allocated based on stocks' membership in benchmark indices, with implications for asset pricing. A large body of literature, starting with Shleifer (1986), shows that stocks added to a benchmark index experience increases in price, volatility, and co-movement.

Whether these asset price distortions affect firm behavior and have implications for allocative efficiency is an open question. In a frictionless market, the value of an investment reflects the present value of expected cash flows, regardless of whom it accrues to or whether they are benchmarked (Modigliani and Miller, 1958). However, an argument that can be traced back to Grossman and Stiglitz (1980) suggests that benchmarking impedes price discovery. As a result, benchmarking may affect firm behavior and lead to misallocation of capital.

This paper studies the causal effects of benchmarking-induced asset price distortions on firm investment. We document a novel channel through which benchmarking affects firm behavior. Our approach uses transitions across Russell index cutoffs to capture exogenous variation in a stock's benchmarking intensity (BMI). We use this variation to provide causal evidence that an increase in a stock's BMI raises its CAPM β. Firm managers perceive this as an increase in their cost of capital, leading to lower investment and capital accumulation. Importantly, we show that these results are not driven by changes in firm fundamentals or cash flow risk. Instead, we argue that these findings are due to firm managers setting discount rates using their stock's CAPM β, without accounting for benchmarking-induced asset price distortions. Our results indicate that benchmarking distortions have first-order implications for investment at the firm- and the aggregate-level, driven by changes in the perceived cost of capital.

We illustrate the main forces of our proposed mechanism with a stylized model. In the model firm managers follow textbook guidance to invest in projects with positive net present value. We compare discount rates for net present value determination under standard textbook assumptions with those under benchmarking-induced distortions. Textbooks advise firm managers to use a discount rate equal to the weighted average cost of capital implied by the firm's CAPM β.[1] However, the presence of benchmarked fund managers introduces wedges into the discount

[1]See, for example, corporate finance textbooks by Brealey, Myers, Allen, and Edmans (2023), Berk and DeMarzo (2023), and Ross, Westerfield, Jaffe, and Jordan (2016). A notable exception is Welch (2022a, Chapter 10) who suggests "benchmarking", i.e. always using a market β of 1 to calculate expected rate of returns.

rate (Kashyap, Kovrijnykh, Li, and Pavlova, 2021). The inelastic demand of benchmarked fund managers raises the price of stocks in the index, but also increases their co-movement with the index. These forces, which are monotonically increasing in BMI, have opposing effects on the discount rate: the increase in the stock price implies a lower discount rate and thus encourages higher investment, whereas an increase in co-movement discourages it. The overall effect of benchmarking on firm investment is ambiguous.

We make a behavioral argument that firm managers may fail to fully account for the distortionary effects of benchmarking. Firm managers who do not internalize the distortionary price effects of benchmarking but continue to set discount rates using their stocks' CAPM βs will perceive an increase in the cost of capital. In this case, increases in a firm's BMI have an unambiguously negative impact on firm investment.

To test the model's predictions, we use the BMI measure developed by Pavlova and Sikorskaya (2023), which assesses the degree to which stocks are included in benchmarked portfolios of institutional investors. BMI measures the total inelastic demand a stock attracts, expressed as a fraction of the stock's market capitalization. We merge the BMI measure with CAPM β estimates from Welch (2022b), accounting data from Compustat, data on managers' perceived costs of capital from Gormsen and Huber (2024), and stock market data from CRSP from 1998 to 2018.

We begin by establishing the causal effect of changes in a stock's BMI on its CAPM β. We use a difference-in-differences design that compares the evolution of CAPM βs between firms experiencing BMI changes around the Russell index reconstitution dates and those that do not. This approach does not require that index inclusion is random or common support in covariate levels across firms. It only requires that treated and control firms would have evolved similarly absent BMI changes. To ensure well-identified estimates, we include high-dimensional fixed effects to remove as much time-varying unobserved heterogeneity as possible.[2]

The difference-in-differences results show that an increase (decrease) in the BMI of a stock by at least 5 percentage points (p.p.) increases (decreases) its CAPM β by 0.16 (-0.14). The treatment effects of index inclusion on CAPM βs increase in BMI. Stocks with a BMI increase of at least 10 p.p. (20 p.p.) subsequently have 0.23 (0.35) higher CAPM βs. These effects are not driven by micro-cap stocks and are also present for stocks with market capitalizations above the 20th percentile of the New York Stock Exchange (NYSE). We confirm that the changes in CAPM βs are not due to changes in firm fundamentals or increased industry-level risk exposure. Instead,

[2] As an alternative strategy, we employ a regression discontinuity design that exploits variation in the BMI of stocks that transition across the Russell index cutoffs. Identification assumes that firms cannot manipulate their market equity rank relative to the index cutoff with precision. Using the Russell 3000 cutoff, we find that the CAPM β of a firm increases (decreases) on average by 0.14 (-0.21) when the firm joins (leaves) the index.

our results indicate that the increased co-movement with the market is explained by institutional ownership, similar to the findings of Antón and Polk (2014).

The resulting increase in a treated firm's perceived equity cost of capital is substantial. Assuming an annual equity risk premium (ERP) of 6%, the baseline results imply an increase of 96 basis points (bps). Firms with BMI increases of more than 10 p.p. experience a 144 bps increase. These increases are significant compared to typical financial price fluctuations, such as long-term interest rate trends and monetary policy shocks. For instance, Bauer and Rudebusch (2020) estimate that the natural real rate of interest declined by approximately 100 bps between 2002 and 2020. Krishnamurthy and Vissing-Jorgensen (2011) find that the Fed's quantitative easing program reduced corporate bond yields by up to 50 bps.

The distortionary effects of BMI increases on CAPM βs persist for at least 7 years. The effects gradually diminish over time but remain statistically significant at each forecast horizon. For a 10 p.p. increase, we find that one year after the initial BMI increase, the equity cost of capital is 150 bps higher (assuming an ERP of 6%). Four years after the shock, it remains 104 bps higher. Even seven years later, we still observe statistically significant effects on the CAPM β which imply a 62 bps higher equity cost of capital.

Using changes in BMI as an instrumental variable (IV) for changes in CAPM βs, we estimate the influence of benchmarking-induced distortions on firm investment. We use Jordà's (2005) local projections (LP) to estimate the effects of an increase in CAPM β on capital allocation over horizons of up to 6 years. The instrument uses the plausibly exogenous variation in BMI to instrument for the endogenous relationship between CAPM β and investment. To ensure that our estimates are well-identified, we use two methods. First, we saturate our LP-IV estimator with high-dimensional fixed effects to remove as much time-varying unobserved heterogeneity as possible. Second, we confirm that our results are robust to inclusion of known predictors of capital accumulation (e.g., Tobin's Q or cash flow) in the LP-IV specifications.

We test whether the IV exclusion restriction holds through a series of additional tests. Concerns may arise that changes in BMI could correlate with changes in risk exposure, debt market access, or governance, potentially violating the exclusion restriction.[3] First, we find that the CAPM βs of peer firms, which typically co-move due to common risk exposure, do not change when a treated firm's BMI changes. Additionally, changes in BMI do not correlate with firm-level risk measures from Hassan, Hollander, Van Lent, and Tahoun (2019), indicating that neither aggregate nor idiosyncratic risk exposure changes with BMI. Second, we test whether higher BMI

[3]Additionally, one might worry that firms self-select into index membership (e.g., by strategically manipulating earnings). Limburg (2024) directly tests this but finds no evidence that firms self-select into index membership.

alleviates financial frictions but find no evidence that text-based measures of financial frictions by Hoberg and Maksimovic (2015) change. Debt financing costs, measured by CDS spreads and their CAPM βs, also remain unchanged when BMI changes, suggesting financial frictions are unaffected by BMI. Third, we assess whether corporate governance changes with index inclusion but find no evidence of changes in governance scores of several providers.[4] The absence of effects on risk exposure, financial frictions, and governance suggests that the fundamentals of firms undergoing changes in BMI remain unchanged. This supports using changes in BMI as valid instruments for changes in a stock's CAPM β and a causal interpretation of our results.

A benchmarking-induced increase in a firm's CAPM β leads to a large and persistent fall in investment. An increase in CAPM β of 15% leads to a cumulative decrease of 10.0% in capital expenditure over six years. Consistent with the slow updating of CAPM βs at the firm-level, the treatment effects are close to zero at short horizons but increase over time and become statistically significant after three years. We find that firms first accumulate cash and subsequently increase payouts to shareholders. The average treated firm's physical and intangible capital stock decline by 7.1% and 8.4% after six years, respectively. The estimated response of the physical capital stock imply user cost of capital elasticities of -1.09 which is close to the theoretical value of unity implied by Cobb-Douglas production technology (Zwick and Mahon, 2017).

Consistent with our mechanism and the observed change in firm investment, we find that increases in BMI predict future increases in managers' perceived cost of capital. Gormsen and Huber (2024) collect self-reported costs of capital estimates from conference calls to measure the perceived cost of capital of managers. We use changes in BMI as an instrumental variable to identify the causal effect of changes in CAPM βs on a firm's perceived costs of capital. The estimates indicate that managers' perceived costs of capital is responsive to benchmarking-induced increases in CAPM β. The IV estimates imply that a two standard deviation increase in a firm's CAPM β (approximately 0.20) raises managers' perceived cost of capital by 80 bps.

Our findings at the firm-level are mirrored at the aggregate-level. From 1998 to 2018, the average BMI increased from 8.16% to 18.32%, while the average CAPM β increased from 0.63 to 1.09. Time-series estimates attribute approximately 50% (or 0.23) of the increase in CAPM β to the rise in average BMI over this time. Remarkably, the time-series estimate of 0.23 is nearly identical to the difference-in-differences estimate for a similar increase in BMI at the firm-level. These results imply that, due to higher levels of benchmarking, the average firm in 2018 perceives its equity cost of capital to be 138 bps higher than in 1998.

[4]Our findings align with Kacperczyk, Sundaresan, and Wang (2021), who also find no evidence that corporate governance changes with increased foreign institutional ownership.

A body of literature finds that U.S. investment has been low in recent decades relative to profitability and valuations (Gutiérrez and Philippon, 2017).[5] According to standard Q-theory, investment should have increased with an increase in Tobin's Q. However, investment rates have been low by historical standards. This has lead researchers to argue that there is "missing investment" which now exceeds 20% of the aggregate capital stock, even after accounting for intangibles and other measurement issues (Gormsen and Huber, 2023).

We argue that increases in aggregate BMI and the resulting distortions in CAPM βs explain over 40% of the missing investment puzzle. We support this argument with industry-level evidence from the NBER-CES manufacturing database. Using IV regressions in long-difference from 2000 to 2016, we measure the effects of CAPM β increases on capital accumulation. The estimates show that higher CAPM βs led to 7.6% lower capital accumulation at the aggregate-level during this period. Gormsen and Huber (2023) calculate that the cumulative investment shortfall in the U.S. from 2002 to 2016 was approximately 17.5%. Our results indicate that the increase in CAPM βs due to benchmarking explains up to 43% of this investment shortfall.

The remainder of this paper is organized as follows. The remainder of this section discusses related literature. Section 2 illustrates our proposed mechanism in a stylized model. Section 3 describes the data. Section 4 establishes a causal link between BMI and CAPM βs. Section 5 tests whether changes in BMI correlate with changes in firm fundamentals. Section 6 shows that BMI-induced changes in CAPM β affect real outcomes at the firm-level. Section 7 tests whether changes in average BMI and CAPM β can explain the aggregate investment shortfall since the early 2000s. Section 8 concludes.

Related Literature This paper contributes to several strands of literature, including the effects of benchmark-linked investing on firm behavior and the broader topic of financial markets and capital (mis-)allocation.

The literature on benchmark-linked investing (see Wurgler, 2010, for a survey) established that stocks appreciate when included in an index and depreciate when deleted (Shleifer, 1986, Harris and Gurel, 1986).[6] Subsequent research found that stock volatility and co-movement with the index increase when a stock is included in an index or ETF and decrease when it is deleted (Vijh, 1994, Barberis, Shleifer, and Wurgler, 2005, Ben-David, Franzoni, and Moussawi, 2018).

Basak and Pavlova (2013) provide a theoretical framework showing that benchmarking fund

[5] Consistent with our findings Gutiérrez and Philippon (2017) find that "within industry-year, the investment gap is driven by firms owned by quasi-indexers" and that these "firms return a disproportionate amount of free cash flows to shareholders". Our findings provide an explanation for why firms owned by quasi-indexers reduce investment: the CAPM β increases with ownership of institutions that are benchmarked to an index.

[6] See also Beneish and Whaley (1996) and Chang, Hong, and Liskovich (2014).

managers leads to asset class effects consistent with observed index inclusion effects. Building on this, Kashyap et al. (2021) derive optimal firm investment policy in the presence of benchmarked fund managers, arguing that managers should internalize the inelastic demand for index stocks and over-invest to maximize firm value. However, due to increased volatility and co-movement (Barberis et al., 2005), the overall effect of benchmarking on investment is ambiguous.[7]

Our findings contribute to Kashyap et al. (2021) in two ways: empirically, we provide causal evidence that increases in BMI lead to lower investment and higher perceived costs of capital. Conceptually, we introduce a novel behavioral argument that reconciles our findings with those of Kashyap et al. (2021). We argue that managers relying on textbook guidance to estimate costs of capital using CAPM fail to internalize benchmarking-induced asset price distortions. This results in an overestimation of the cost of capital and a sub-optimal decline in investment, ultimately destroying shareholder value.

Our findings link to a broader literature on how benchmark-linked investing affects firm decision-making. Consistent with our results, Asker, Farre-Mensa, and Ljungqvist (2014) find that public firms invest less than private firms. On benchmark-linked investing, Appel, Gormley, and Keim (2016), Edmans and Holderness (2017), Bebchuk, Cohen, and Hirst (2017), Heath, Macciocchi, Michaely, and Ringgenberg (2021), and Lewellen and Lewellen (2022) examine its effects on firm governance, while Azar, Schmalz, and Tecu (2018) study the anti-competitive implications of common ownership.

Additionally, our paper contributes to the literature on how firm managers determine discount rates and the role of the textbook CAPM. The Duke CFO survey (Graham, 2022) finds that the CAPM is the leading method for determining discount rates. Further evidence from Gormsen and Huber (2024) and Dessaint, Olivier, Otto, and Thesmar (2020) shows that practitioners rely on the CAPM to estimate discount rates from earnings calls and M&A transactions. Berk and Van Binsbergen (2016) infer that the CAPM closely aligns with the model investment fund managers use for capital allocation decisions. Andrei, Cujean, and Wilson (2023) argue that the empirical failure of the CAPM, despite its widespread practical use, is due to variations in expected returns over time and across investors, leading to mismeasured CAPM βs and perceived pricing of non-existent factors.

[7] See appendix D of Kashyap et al. (2021) for further details.

2 A stylized model that highlights the mechanism

Every standard corporate finance textbook instructs firm managers to create and implement investment policies that maximize firm value. The canonical textbook guidance is to maximize net present value (NPV).[8] Firm managers need two key components to calculate NPV: the expected cash flows and the discount rate to convert future cash flows to present value. While most textbooks provide little guidance on estimating the former, they typically recommend using the weighted average cost of capital (WACC) as discount rate and to estimate the CAPM β to determine the equity cost of capital. The following stylized model highlights the key mechanisms and distortions to the discount rate introduced by benchmarking of asset managers.

Textbook investment policy The value of a firm $V_{i,t}$ is determined by the net present value of its expected future cash flows $\{CF_{i,t+h}\}_{h=1}^{\infty}$ discounted at the firm-specific discount rate $\{R_{i,t\to t+h}\}_{h=1}^{\infty}$:

$$V_{i,t} = \mathbb{E}_t \left[\sum_{h=1}^{\infty} \frac{CF_{i,t+h}}{R_{i,t\to t+h}} \right]. \tag{1}$$

The discount rate $R_{i,t\to t+h}$ equals the firm's weighted average cost of capital, determined by exposure to aggregate risk[9] of the cash flows generated by the firm's assets β_i^A and the yield curve of risk-free rates $\{R_{t+h}^f\}_{h=1}^{\infty}$.

$$R_{i,t\to t+h} = \prod_{j=1}^{h} \left(R_{t+j}^f + \beta_i^A ERP_{t+j} \right) \tag{2}$$

Assuming for simplicity that firm leverage remains constant over time and is sufficiently low to not create default risk, the aggregate risk exposure of the firm's cash flows is proportional to exposure of the firm's equity to the equity risk premium (ERP_t):

$$\beta_i^A = \frac{\beta_i^E}{1 + (1-\tau)\frac{D_i}{E_i}} \tag{3}$$

As such, it can be directly inferred from the empirical market beta of a firm's equity $\widehat{\beta}_i^E = \frac{\widehat{\text{Cov}}(r_i, r_m)}{\widehat{\text{Var}}(r_m)}$. A firm manager considering a firm-typical project with cost $C_{i,t}$ and future cash flows

[8] See Ben-David and Chinco (2023) for a model in which managers maximize earnings-per-share instead.

[9] We abstract away from variation in term structure of equity yields for simplicity. Early empirical studies indicated a declining term structure of equity yields (Van Binsbergen, Hueskes, Koijen, and Vrugt, 2013), while more recent work suggests equity yields are either stable (Giglio, Kelly, and Kozak, 2023) or increase with horizon (Bansal, Miller, Song, and Yaron, 2021).

$\{y_{i,t+h}\}_{h=1}^{\infty}$ should invest in the project if it increases firm value, that is if

$$\mathbb{E}_t\left[\sum_{h=1}^{\infty}\frac{y_{i,t+h}}{\widehat{R}_{i,t+h}}\right]=\mathbb{E}_t\left[\sum_{h=1}^{\infty}\frac{y_{it+h}}{\prod_{j=1}^{h}\left(R^f_{t+j}+\frac{\widehat{\beta}^E_i}{1+(1-\tau)\frac{D_i}{E_i}}ERP_{t+j}\right)}\right]>C_{i,t}, \tag{4}$$

with $\widehat{R}_{i,t}$ denoting the discount rate that results from plugging the empirical counterpart of (3) into (2).

Benchmarking distortions The literature has established that the presence of benchmarked fund managers distorts asset prices (see e.g., Basak and Pavlova 2013 and Buffa, Vayanos, and Woolley 2022). Kashyap et al. (2021) argue that benchmarked fund managers' inelastic demand for index constituent stocks increases their stock price and thereby lowers the discount rate for index constituent firms. However, index membership also induces excess co-movement (Barberis et al., 2005) between index constituent stocks that is unrelated to the aggregate risk exposure of the firm's cash flows.[10] Excess co-movement leads to an increase in β^E_i and thus increases the discount rate proportional to the ERP.

We illustrate these two opposing effects in reduced form by postulating two discount rate wedges as functions of a stock's BMI. The price pressure from index inclusion (Shleifer, 1986) reduces the effective discount rate by $\Delta_I(BMI_{i,t})$. At the same time, the equity market β^E increases to $\widetilde{\beta}^E_{i,t}=\beta^E_i+\Delta_\beta(BMI_{i,t})$. Both $\Delta_I(BMI_{i,t})$ and $\Delta_\beta(BMI_{i,t})$ monotonically increase in $BMI_{i,t}$ and satisfy $\Delta_I(0)=\Delta_\beta(0)=0$.

The discount rate, adjusted for benchmarking distortions, that maximizes firm value is

$$\widetilde{R}_{i,t\to t+h}=\prod_{j=1}^{h}\left(R^f_{t+j}+\frac{\beta^E_i+\Delta_\beta(BMI_{i,t})}{1+(1-\tau)\frac{D_i}{E_i}}ERP_{t+j}-\Delta_I(BMI_{i,t})\right). \tag{5}$$

The total effect of both discount rate wedges is ambiguous and depends on the functional forms of $\Delta_I(BMI_{i,t})$ and $\Delta_\beta(BMI_{i,t})$. Empirically, positive price effects at index inclusion (see Chang et al., 2014 or Pavlova and Sikorskaya, 2023) suggest that $\Delta_I(BMI_{i,t})$ dominates in the short run.

CAPM investment policy with benchmarked fund managers Whether the benchmarking distortions to the discount rate in Eq. (5) cause firm managers to invest more or less is ambiguous. The level-shift in discount rates by $\Delta_I(BMI_{i,t})$ (in blue) incentivizes higher investment, whereas the increase in CAPM β, Δ_β (in red), incentivizes lower investment.

[10] See also Appendix D of Kashyap et al. (2021) for further details.

What if firm managers do not internalize the inelastic demand of benchmarked fund managers for index constituent stocks but use the cost of capital implied by their CAPM β to evaluate investment opportunities as in Eq. (4)? In this case, the presence of benchmarked fund managers has an unambiguously negative effect on investment for index constituent firms. Firm managers observe an increase in their equity-β from $\widehat{\beta}_i^E$ to $\widetilde{\beta}_i^E = \widehat{\beta}_i^E + \Delta_\beta(BMI_{i,t})$ and infer an increase in their cost of capital. Firm managers now invest only in projects that satisfy

$$\mathbb{E}_t\left[\sum_{h=1}^{\infty}\frac{y_{i,t+h}}{\widetilde{\widetilde{R}}_{i,t+h}}\right] = \mathbb{E}_t\left[\sum_{h=1}^{\infty}\frac{y_{i,t+h}}{\prod_{j=1}^{h}\left(R^f_{t+j} + \frac{\widehat{\beta}_i^E + \Delta_\beta(BMI_{i,t})}{1+(1-\tau)\frac{D_i}{E_i}}ERP_{t+j}\right)}\right] > C_{i,t}. \tag{6}$$

All else being equal, a firm within the index invests less than the same firm outside the index.

Testable hypothesis Our proposed mechanisms rests on the behavioral assumption that firm managers do not internalize the total effect of benchmarking-induced discount rate distortions. Instead, managers follow canonical textbook guidance to form a weighted average cost of capital using their firm's empirical CAPM βs. Empirical CAPM βs, however, are contaminated by benchmarking-induced co-movement and will thus lead index constituent firms to under-invest.

We empirically validate our mechanism by presenting evidence that supports three testable hypotheses directly derived from it. All else equal,

(i) there is a monotonic positive relationship between changes in firm BMI and changes in β^E,
(ii) an increase in firm BMI increases the firm's perceived cost of capital,
(iii) and leads to a decline in investment.

3 Data and sample

We use four main sources in our empirical analysis: (1) the measure of benchmarking intensity developed by Pavlova and Sikorskaya (2023), (2) estimates of firm CAPM βs from Welch (2022b), (3) firm-level data from S&P's Compustat North America Fundamentals, and (4) additional firm-level variables from various sources.

Benchmarking Intensity We use the benchmarking intensity measure developed by Pavlova and Sikorskaya (2023) from 2001 to 2018. Benchmarking intensity (BMI) for stock i in month t is

Table 1: Summary Statistics of Matched BMI-CAPM Sample

	Mean	SD	Min	P5	P10	Median	P90	P95	Max	N
CAPM β (Welch, 2022)	0.90	0.49	-2.00	0.13	0.25	0.90	1.54	1.73	3.78	976,577
Δ CAPM β	0.03	0.26	-1.67	-0.40	-0.28	0.03	0.34	0.45	1.88	54,474
BMI in May	0.16	0.09	0.00	0.00	0.00	0.18	0.26	0.28	3.02	55,872
BMI in June	0.16	0.09	0.00	0.00	0.00	0.18	0.26	0.28	1.08	55,872
Δ BMI	0.00	0.04	-0.56	-0.04	-0.02	0.00	0.03	0.05	0.42	55,871
Market Capitalization (in $m)	3,972	19,836	1	12	23	327	6,189	14,841	1,099,436	976,577
Shares Outstanding (in 1000s)	106.86	432.80	0.18	3.08	4.92	28.48	181.76	363.34	29,206.40	976,577
Trading Volume (in 100,000s)	19.20	103.29	0.00	0.00	0.00	2.00	40.00	78.00	20,124	976,577

Notes: Monthly sample from 2001-01 to 2019-06. Δ CAPM β is the difference between average CAPM β in the first and last quarter of a year.

defined as

$$\text{BMI}_{i,t} = \sum_{j=1}^{J} \frac{\lambda_{j,t}\omega_{i,j,t}}{\text{Market Value}_{i,t}} = \sum_{j=1}^{J} \frac{\lambda_{j,t}\mathbb{1}\{\text{Stock is in Index}\}_{i,t}}{\text{Market Value of all Stocks in Index}_{j,t}} \tag{7}$$

where $\lambda_{j,t}$ is the assets under management (AUM) of mutual funds and ETFs benchmarked to index j and $\omega_{i,j,t} = \frac{\text{Market Value}_{i,t}\mathbb{1}\{\text{Stock is in Index}\}_{i,t}}{\text{Market Value of all Stocks in Index}_{j,t}}$ is the stock weight in value-weighted index j.

Similar to Pavlova and Sikorskaya (2023), we use changes in BMI between the rank day of Russell indices in May and the reconstitution day in June as an instrumental variable for changes in institutional ownership. Changes in BMI satisfy the relevance condition because they predict how benchmarked investors rebalance their portfolios after a Russell index reconstitution. The exclusion restriction requires that index membership is exogenous. The literature on index inclusion effects argues that after controlling for factors that determine index inclusion, most importantly the ranking variable (market value) that Russell uses for index assignment at the end of May, the index membership dummy is exogenous.

CAPM βs We use monthly estimates of the slope winsorized and exponentially weighted CAPM βs of Welch (2022b). The β estimator first winsorizes a firm's daily stock return at −2 and +4 times the contemporaneous market return, and then estimates an exponentially weighting least squares regression of the stock's (winsorized) excess return on the market excess return. The half-life of the exponential decay is 90 days. Welch (2022b) shows that this estimator outperforms other well known β estimators like Bloomberg β or the Vasicek (1973) β in predicting future market βs. We focus on common equities that are traded on either the NYSE, AMEX, or NASDAQ, and exclude ADRs, REITs, and ETFs. Table 1 reports descriptive statistics for the matched BMI-CAPM dataset covering 2001-1 to 2019-06.

Firm-level data We use annual data for publicly listed companies incorporated and located in the U.S. from Compustat for the sample period from 2000 to 2018. We exclude financial firms (SIC codes 6000-6999) and firms in regulated industries (4900-4999), as well as firms with less than \$50m in total assets or less than \$10m sales (in 2017 dollars). Firms must have at least five years of consecutive data. We winsorize the data at the 2.5% and 97.5% level to account for outliers.

Additional firm data We obtain stock returns and market data from the Center for Research in Security Prices (CRSP). Data on firm managers' perceived cost of capital is sourced from Gormsen and Huber (2024), who hand-collect these from earnings calls. Measures of firm-level intangible capital are from Peters and Taylor (2017). Executive compensation peer group information is collected from Institutional Shareholder Services (ISS). We use firm-level risk measures from Hassan et al. (2019) and data on financial frictions from Hoberg and Maksimovic (2015) and Linn and Weagley (2021). Governance scores are obtained from Sustainalytics, Refinitiv, and S&P.

4 Changes in a firm's CAPM β due to Benchmarking

We measure the causal effect of changes in a firm's BMI on its CAPM β using a difference-in-differences design. We compare the evolution of CAPM βs of (treated) stocks that experience BMI changes around Russell index reconstitution dates to (control) stocks that do not. Our treatment group consists of stocks that experience changes in BMI greater than ± 5 p.p.. The control group consists of stocks that experience changes in BMI smaller than ± 1.5 p.p..

The difference-in-differences results show that changes in a stock's BMI causally affect its CAPM β. This effect is symmetric and monotonically increasing in treatment intensity. A stock whose BMI increases (decreases) by at least 5 p.p. experiences an increase (decrease) in CAPM β by 0.16 (-0.14) relative to the control group. A stock whose BMI increases by 10 p.p. to 20 p.p. (more than 20 p.p.) experiences a 0.24 (0.35) increase in CAPM β. These effects are large and economically meaningful. The baseline treatment effect of an increase in BMI greater than 5 p.p. translates into a 96 bps increase in the equity cost of capital (assuming a 6% ERP). An increase in BMI by 10 p.p. to 20 p.p. (greater than 20 p.p.) translates into increase of 132 bps (210 bps).

In Appendix A.3 and A.4 we find qualitatively and quantitatively similar results using regression discontinuity designs (RDD) strategy that exploits BMI discontinuities for stocks transitioning across Russell index cutoffs.

4.1 Difference-in-differences strategy

We analyze the effect of an increase in BMI on a firm's CAPM β by estimating a series of difference-in-differences specifications of the form:

$$\text{CAPM } \beta_{i,t} = \delta \text{ Treated}_i \times \text{Post}_{t>\text{May}} + \theta_i + \theta_{t,s} + \varepsilon_{i,t} \tag{8}$$

where Treated_i is an indicator variable for whether firm i's BMI changed by more than ± 5 p.p.. The coefficient of interest, δ, summarizes the treatment effect of a change in BMI on a firm's CAPM β.

The inclusion of firm fixed effects θ_i ensures that we remove time-invariant heterogeneity across firms and, in particular, accounts for possible ex-ante differences in characteristics between treated and control firms. Size-by-month or liquidity-by-month fixed effects $\theta_{s,t}$ restrict the identifying variation to comparing firms within the same size or liquidity decile each period. This controls for time-varying unobserved heterogeneity across firm size and liquidity, which may be correlated with a firm's CAPM β. We drop stocks with market values below the fifth percentile of stocks traded on the NYSE to ensure that our results are not driven by micro-cap stocks.[11]

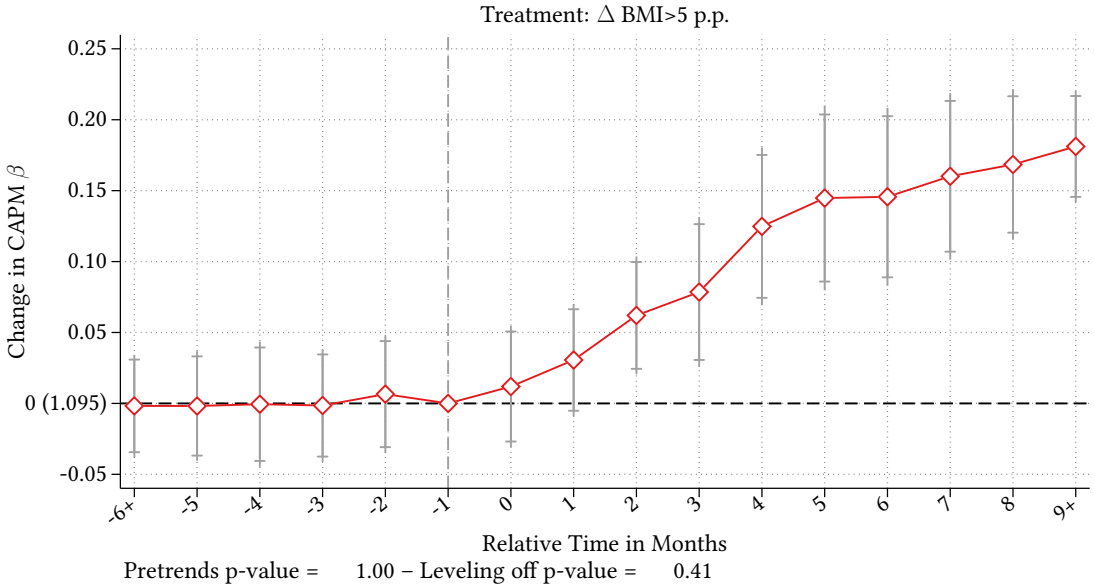
The Russell indices reconstitute each year, and identification in (8) thus exhibits staggered treatment timing with potential misleading estimates when the treatment effect is heterogeneous between cohorts or over time (De Chaisemartin and d'Haultfoeuille, 2023). We follow the suggestion of Baker, Larcker, and Wang (2022) to stack the yearly cohorts and include cohort by firm and cohort by time fixed effects but suppress cohort subscripts for brevity.[12]

Identifying assumptions and threats to identification. Our identification strategy relies on the following assumption: conditional on the rich set of fixed effects and other control variables, firms that experience changes in BMI are not differentially exposed to unobservable shocks that are correlated with BMI. The identification assumption does not require random assignment of index inclusion, nor does it require that firms have similar characteristics in levels. Rather, we rely on the parallel trends assumption that outcomes for treated and control firms would have trended similarly absent an increase in BMI.

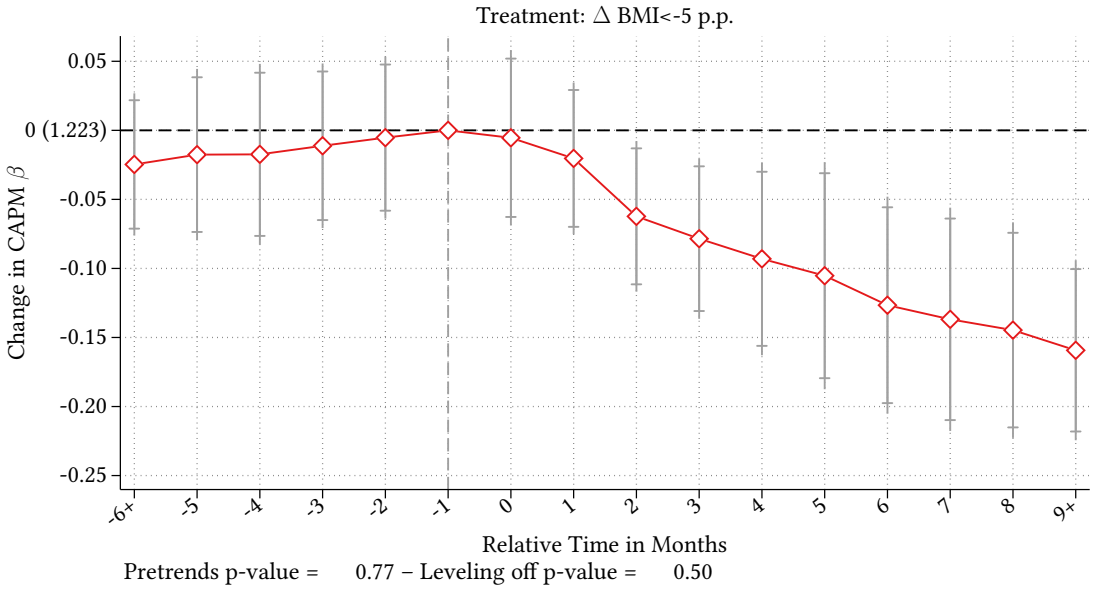
Although, by definition, our identifying assumption is untestable, we provide supporting evidence of parallel pre-treatment trends in Figure 1a and 1b. We additionally perform a placebo test

[11] We obtain similar results when using the 10th or 20th percentile of the NYSE as a cutoff (see Appendix Table A1).
[12] See, e.g., Gormley and Matsa (2011), Cengiz, Dube, Lindner, and Zipperer (2019) and Deshpande and Li (2019) for similar implementations of stacked difference-in-differences designs.

Figure 1: Difference-in-differences Event Study



(a) Treated stocks with an increase in Δ BMI > 5 p.p. relative to control group.



(b) Treated stocks with a decrease in Δ BMI < -5 p.p. relative to control group.

Notes: Change in CAPM β of treated stocks with an increase or decrease in BMI of at least 5 p.p. relative to a control group, where $|\Delta \text{ BMI}| < 1.5$ p.p.. Pointwise confidence intervals (99%) and sup-t confidence bands based on double clustered standard errors. Values in parentheses on the Y-axis show the average CAPM β before treatment.

Table 2: Causal Effects of Changes in Benchmarking Intensity on CAPM βs

Treatment Group:	Δ BMI>5p.p.				Δ BMI<-5 p.p.			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated × Post	0.158***	0.164***	0.161***	0.160***	-0.143***	-0.133***	-0.135***	-0.136***
	(0.014)	(0.014)	(0.014)	(0.014)	(0.021)	(0.021)	(0.020)	(0.020)
Firm × Cohort	✓	✓	✓	✓	✓	✓		
Time × Cohort	✓				✓			
Size Decile × Time × Cohort		✓				✓		
Volume Decile × Time × Cohort			✓				✓	
Shrs. Out. Decile × Time × Cohort				✓				✓
Observations	316,883	316,848	316,762	316,695	308,973	308,927	308,855	308,782

Notes: Standard errors in parentheses are double clustered at firm and year-month level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001. Sample from 2000-01 to 2019-06. Treated×Post is the average of post-treatment coefficients after 4 months (to account for the expanding-window estimation of CAPM β). The control group are firms whose BMI changes less by than 1.5 p.p. in absolute value.

that compares firms with BMI increases between 0 p.p. and 1 p.p. and firms with BMI decreases between 0 p.p. and -1 p.p. in Figure A1. As expected, this placebo test finds no effects.

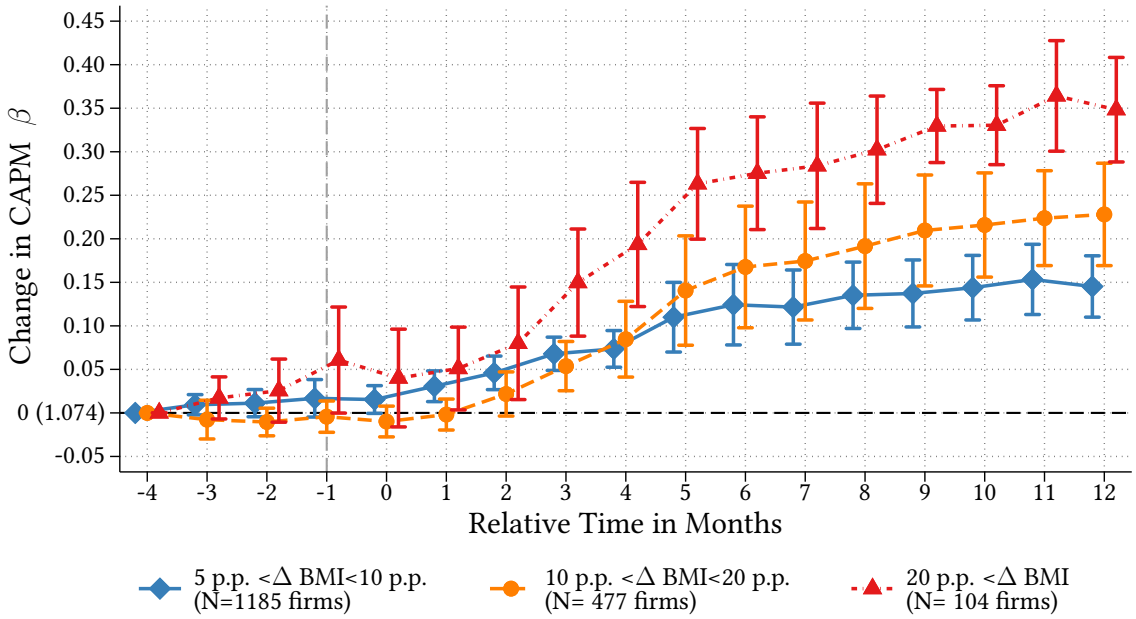
Results Figures 1a and 1b show the event study coefficients of our difference-in-differences estimation for treatments defined as an increase or decrease in BMI of 5 p.p., respectively. We normalize the dynamic treatment effect to zero in May and estimate dynamic treatment effects for the period from 9 months before to 12 months after the change in BMI.

Several facts are worth noting. First, the CAPM βs of treated and control firms evolve in close parallel before index reconstitution. This supports the identification assumption and causal interpretation of treatment effects. The p-values of the Wald-style pre-trend test suggested by Freyaldenhoven, Hansen, Pérez, and Shapiro (2021) are 1.00 and 0.77, respectively. Second, for both increases and decreases in BMI, CAPM βs start to diverge after index reconstitution. Note that the CAPM βs are estimated on an expanding-window using exponentially weighted least squares regressions with a half-life of 90 trading days. This implies that the smooth event-time trend visible in Figure 1 is to a large extent mechanical, as older information receives exponentially smaller weights. This further implies that the treatment effect of an increase in BMI on a firm's CAPM β is likely immediate, but our measurement only picks up on this when older information is sufficiently down-weighted. Third, treatment effects for both increases and decreases in BMI by at least 5 p.p. have similar magnitudes of opposite sign, suggesting that treatment effects are linear in BMI.

Table 2 reports the average estimated post-shock coefficient.[13] The coefficient estimate of the interaction $\text{Treated}_i \times \text{Post}_{t>\text{May}}$ is positive for increases in BMI and negative for decreases in BMI.

[13] We exclude the coefficients for $0 \leq t < 5$ from the calculation of the static difference-in-differences specification, as the CAPM βs are estimated using expanding windows and thus by construction have a dynamic component.

Figure 2: Difference-in-differences Event Study for Different Treatment Intensities



Note: Estimated using local projections difference-in-differences of Dube et al. (2023).

Notes: Change in CAPM β of treated stocks for different treatment intensities relative to control group where $|\Delta$ BMI$|$<1.5 p.p.. Pointwise confidence intervals (90%) based on double clustered standard errors. Values in parentheses on the Y-axis indicate the median CAPM β before treatment for all treatment intensities.

The estimated treatment effects are always statistically significant at the 0.1% level. Columns (1) and (5) show the results with firm and year-month fixed effects. Subsequent columns add more granular year-month by size, volume, or shares outstanding fixed effects. In particular, these fixed effects control for the possibility that treated firms might differ in terms of size or liquidity from control firms. The estimated effects of changes in BMI are stable across the different sets of fixed effects. Column (2) implies that an increase in BMI by at least 5 p.p. increases a firm's CAPM β by 0.16, while Column (6) implies that a decrease in BMI by at least 5 p.p. decreases a firm's CAPM β by -0.14. These effects are large and economically meaningful. An increase of 0.16 in CAPM β implies a 96 bps increase in the equity cost of capital (assuming an ERP of 6%).[14]

However, the average treatment effect estimates in Table 2 mask substantial heterogeneity. Figure 2 plots the event-time coefficients for three different treatment intensities. Treatment levels are defined in increasing order of treatment intensity as changes in BMI by 5 p.p. to 10 p.p., 10 p.p. to 20 p.p., and greater than 20 p.p.. The estimated treatment effect is 0.15 for the lowest treatment intensity, 0.24 for the medium treatment intensity, and 0.35 for the highest treatment

[14] Jacobs and Shivdasani (2012) report that 49% of surveyed managers use an ERP between 5% and 6%.

Table 3: Persistence Effects of a 10 p.p. increase in Benchmarking Intensity on CAPM β

	Dependent Variable: CAPM β in t+h Years After Change in BMI:						
	t+1	t+2	t+3	t+4	t+5	t+6	t+7
Δ BMI$_t$	0.252***	0.225***	0.191***	0.173***	0.146***	0.128***	0.104***
(scaled to 10 p.p. increase)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)
Controls	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.21	0.19	0.15	0.12	0.10	0.08	0.07
Observations	61,127	57,195	53,535	47,777	42,586	37,886	33,565

Notes: This table reports coefficient estimates for γ_1^h estimates from regressions: CAPM $\beta_{i,t+h} = \gamma_0^h + \gamma_1^h \Delta\text{BMI}_{i,t} + \zeta^h X_{i,t} + \varepsilon_{i,t+h}$ for $h = 1, \ldots, 7$. CAPM β_{t+h} is average of monthly estimates in year $t + h$. Estimates are scaled to a 10 p.p. increase in BMI. Yearly sample from 1998 to 2022. All variables are winsorized at the 2% and 98% level to remove outliers. Standard errors in parentheses are clustered at firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

intensity after 12 months. The treatment effects are statistically significant at the 0.1% level for all treatment intensities. For firms in the highest treatment intensity group, the treatment effect is 0.35, which is equivalent to a 210 bps increase in the perceived equity cost of capital.

Persistence Our difference-in-difference results show that the distortionary effects of BMI increases persist for at least 12 months after treatment. However, CAPM βs tend to revert to 1 in the long run (Blume, 1975). If the distortionary effects of benchmarking on CAPM βs fade after a couple of years, we would not expect any long-term impact on investment. We therefore test whether changes in the CAPM βs persistent over longer horizons using the following specification:

$$\text{CAPM } \beta_{i,t+h} = \gamma_0^h + \gamma_1^h \Delta\text{BMI}_{i,t} + \zeta^h X_{i,t} + \varepsilon_{i,t+h}, \quad h = 1, \ldots, 7 \tag{9}$$

The coefficients of interest, γ_1^h, summarize the long-term effect of an BMI increase on a firm's CAPM β after h years. The vector of controls $X_{i,t}$ contains year fixed effects and the lagged level of BMI.

Table 3 shows coefficient estimates for γ_1^h of Eq. (9) for a forecast horizon of up to 7 years. For ease of interpretation, the estimates are scaled to a 10 p.p. increase in BMI. Several facts are worth noting. First, the distortionary effects of increases in BMI persist for up to 7 years. At each forecast horizon we find statistically significant (p<0.001) positive effects of an increase in BMI on CAPM βs. Second, the effects diminish gradually over time.[15] The estimates imply economically significant magnitudes. One year after the initial BMI increase, the equity cost of capital is 151

[15] In unreported regressions, we find that the relationship is more stable when we restrict the sample to 2010 to 2018, likely explained by the increasing time trend of benchmarking intensity.

bps higher (assuming an ERP of 6%). Four years after the shock, it remains 104 bps higher. Even seven years later, we still observe effects on the CAPM β, implying a 62 bps higher equity cost of capital.

Alternative estimators of the CAPM β Our results show that changes in BMI cause changes in CAPM β. However, the fundamental CAPM β of a stock is unobservable and must be estimated. We use the estimator proposed by Welch (2022b) due to its superior performance in predicting future CAPM βs out of sample. Nevertheless, there may be concerns that our choice of estimator influences our results. To verify that our results are not driven by the choice of estimator, we use the following reduced form specification for a set of alternative CAPM β estimators:

$$\Delta\beta_{i,t}^{ALT} = \alpha_i + \alpha_t + \gamma\Delta\text{BMI}_{i,t} + \varepsilon_{i,t} \tag{10}$$

Appendix Table A2 shows reduced form regression results using different estimators for the unknown CAPM β. We compare the usual OLS estimator with alternative estimators proposed by Blume (1975) (the "Bloomberg β"), Dimson (1979), and Welch (2022b). We also report estimates of β for the ten largest stocks by market capitalization, and correlation with the market and idiosyncratic volatility. For each estimator, we provide expanding-window (using exponentially decaying weights) estimates in Panel A and 2-year rolling-window (using equal weights) estimates in Panel B.

We find that *all* considered CAPM β estimators show a statistically significant (p<0.001) association with changes in BMI. Estimates using rolling-windows with equal weighting are smaller in magnitude but also highly statistically significant. Changes in CAPM β occur due to changes in market correlation, not because of increases in idiosyncratic volatility. This aligns with Antón and Polk (2014), who show that mutual funds' common ownership increases the return correlation of equities.

5 Additional tests and instrument validity

Our first main result shows that increases in a stock's BMI causally increase its CAPM β. We use changes in BMI between the Russell ranking day in May and the reconstitution day in June as an instrumental variable to exploit this effect. Concerns may arise that other factors, such as risk exposure, access to debt markets, or governance, could change alongside CAPM β with index membership changes, potentially violating the exclusion restriction. However, our additional tests show no evidence that changes in BMI correlate with changes in risk, financial frictions, or

governance.

5.1 Changes in BMI and measures of risk exposure

A potential concern with our IV identification strategy is that changes in BMI could correlate with changes in exposure to aggregate or idiosyncratic risk. Firm-level exposure to aggregate risk is influenced by an industry's aggregate risk (Karolyi, 1992) and by firm fundamentals (e.g., as in the model of Gomes, Kogan, and Zhang, 2003). We test whether the aggregate risk exposure of treated firms changes by estimating whether the CAPM β of comparable peer firms changes when a firm's BMI changes. We also test whether measures of idiosyncratic firm-level risk exposure change with BMI. However, we find no evidence that changes in BMI correlate with changes in aggregate or idiosyncratic risk exposure.

Changes in the CAPM β of peer firms We collect information about a firm's executive compensation peer group from ISS. For each firm, we randomly select three peer firms and test whether the firm's change in BMI correlates with changes in the CAPM β of peer firms. To avoid confounding our estimates, we exclude peer firms that also experience a change in BMI. Appendix Table A3 presents the results of this placebo test using a firm's peers. Column (1) reports the coefficient from a regression of changes in the firm's CAPM β on changes in the CAPM β of peer firms. The coefficient is positive and statistically significant (p<0.001). The point estimate of 0.232 indicates that a substantial part of the variation in CAPM βs is explained by common exposure to aggregate risk among peer firms. Column (2) reports the reduced form analog of the causal estimates, showing that changes in a firm's BMI are significant drivers of changes in its CAPM β. Columns (3) to (6) test whether changes in a firm's BMI correlate with changes in peers' CAPM βs. Across all four columns, the coefficients are close to zero and are not statistically different from zero.

Firm-level risk measures We analyze six firm-level risk measures derived from earnings calls by Hassan et al. (2019): the overall risk exposure of firms, exposure to overall political risk, and exposure to political risk stemming from economic policy, security policy, technological policy, and trade policy. Appendix Table A4 reports estimates of OLS regressions of changes in firm-level risk measures on changes in CAPM β and changes in BMI. Two things are worth noting. First, changes in the CAPM β correlate with changes in the firm-level risk measures. Four of six firm-level risk-measure show a statistically significant positive relationship with changes in the CAPM β of firms. Second, changes in BMI do not correlate with changes in firm-level risk measures. The estimated coefficients across all risk measures are close to zero and are not statistically significant.

5.2 Changes in BMI and measures of financial constraints

Changes in BMI could also correlate with changes in financial constraints, potentially violating the exclusion restriction of our IV strategy. We test this by examining the correlation between changes in a firm's BMI and text-based measures of financial constraints from Hoberg and Maksimovic (2015) and CDS spreads. Both tests directly address the validity of our instrumental variable design. The exclusion restriction would be violated if changes in BMI correlated with changes in the firm's financing costs due to factors other than changes in CAPM β. However, we find no evidence that changes in BMI correlate with changes in financial constraints or debt financing cost.

Text-based measures of financial constraints We collect text-based measures of financial constraints from Hoberg and Maksimovic (2015) and Linn and Weagley (2021). These measures capture the extent to which firms face financial constraints and are likely to constrain investment based on the text of their annual reports. Appendix Table A6 reports estimates of OLS regressions of changes in the firm's financial constraints on changes in the BMI of a firm. The estimated coefficients of BMI are close to zero and not statistically significant across all measures. Importantly, Column (1) of Appendix Table A6 shows that changes in BMI do not correlate with firm statements about delaying investments.

Changes in CDS spreads and CDS CAPM β We collect CDS spreads for senior unsecured debt with tenor of 5 year (no restructuring) from 2010[16] to 2019. CDS CAPM βs are calculated on daily data using the estimator of Welch (2022b). We calculate changes in a firm's CDS spreads and firm's CAPM β of CDS spreads as the difference between the average of daily observations in the first and last quarter of a year. Appendix Table A5 reports estimates of OLS regressions of changes in CDS spreads and changes in the CDS CAPM β on changes in the BMI of a firm. We find no evidence that changes in the BMI predict changes in firm CDS spreads. Similarly, changes in BMI are not correlate with changes in firm debt exposure to aggregate risk as measured by the CDS CAPM βs. The estimated coefficients of BMI are close to zero and are not statistically significant.

5.3 Changes in BMI and measures of corporate governance

An alternative channel through which an increase in BMI and associated changes in institutional ownership could affect investment is improved corporate governance through better monitoring

[16]The International Swaps and Derivatives Association (ISDA) introduced a number of changes in its "Big Bang" of April 2009. We therefore focus on the after-reform period in order to have a coherent sample.

(Appel et al., 2016, Aghion, Van Reenen, and Zingales, 2013). However, increased passive ownership may also decrease monitoring incentives, as in the model of Bebchuk and Hirst (2019). We test whether measures of corporate governance change with changes in BMI but find no evidence of such an effect.

Governance Scores We obtain governance and ESG scores from S&P, Sustainalytics, and Refinitiv and test whether changes in BMI correlate with changes in those scores. Appendix Table A7 reports estimates of OLS regressions of changes in governance and ESG scores on changes in the BMI of a firm. The estimated coefficients are close to zero and are not statistically significant.

6 Effects of CAPM β distortions on capital accumulation

Our second set of results estimates how firms react to changes in their CAPM β induced by changes in BMI. For a firm manager who follows textbook guidance to set investment policies using the CAPM, an increase in a firm's CAPM β increases the user cost of capital and should lead to a decrease in investment. We therefore test whether changes in CAPM β affect firm outcomes like capital expenditure, physical and intangible capital stocks, cash holdings, and payouts to shareholders.

The results show that benchmarking-induced increases in the CAPM β lead to a large and persistent decline in investment. Capital expenditure declines by a cumulative of 10.0% over six years. The average firm's physical and intangible capital stock is 7.1% and 8.4% lower after six years, respectively. Firms initially accumulate cash and subsequently increase payouts to shareholders. These findings are robust to inclusion of other known predictors of investment, such as firm size, cash flow, and Tobin's Q.

Consistent with our mechanism, managers' perceived costs of capital respond to benchmarking-induced increases in CAPM β. Using data from Gormsen and Huber (2024), we show that increases in BMI predict future increases in managers' perceived cost of capital. We use changes in BMI as an instrumental variable to identify the causal effect of changes in CAPM β on a a firm's perceived cost of capital. The IV estimates imply that a two standard deviation change in a firm's CAPM β (≈ 0.20) increase the managers' perceived cost of capital by 80 bps.

6.1 Empirical Strategy

We use an instrumental variable (IV) local projections (LP) strategy to forecast the effects of a change in CAPM β on capital allocation over horizons of up to 6 years.[17] We instrument changes in CAPM β with changes in BMI, in an effort to identify causal effects. Pavlova and Sikorskaya (2023) argue that changes in BMI between the rank day of Russell indices in May and the reconstitution day in June serve as instruments for changes in institutional ownership. Our results in Section 4 show that changes in the level of institutional ownership cause changes in the firms' CAPM βs. Section 5 shows that these changes in CAPM β are orthogonal to aggregate risk exposure and fundamentals of the firm, thus acting as exogenous shock to the firm's cost of capital.

To analyze the effect of BMI-induced changes in a firm's CAPM β on real outcomes, we estimate a series of local projection instrumental variable regressions of the following form:

$$\Delta\text{CAPM } \beta_{i,t} = \delta_i + \delta_{j,t} + \theta \Delta\text{BMI}_{i,t} + \zeta X_{i,t} + \epsilon_{i,t} \tag{11}$$

$$\log\left(Y_{i,t+h}\right) - \log\left(Y_{i,t-1}\right) = \alpha_i^h + \alpha_{j,t}^h + \gamma^h \Delta\widehat{\text{CAPM}}\, \beta_{i,t} + \xi^h X_{i,t} + \varepsilon_{i,t+h} \tag{12}$$

for firm i in industry j in calendar year $t+h$. The coefficients of interests, γ^h, provide cumulative local average treatment effects in % after $h = 0, 1, \ldots, 6$ years.

We remove time-invariant heterogeneity across firms by including firm-fixed effects α_i and δ_i in both first and second stage. We additionally include (3-digit SIC) industry-by-year-by-total-asset quintile fixed effects $\alpha_{j,t}$ and $\delta_{j,t}$ to control for time-varying unobserved heterogeneity across industries, such as differences in industry-level business cycles, which may be correlated with firm outcomes. The use of industry-by-year-by-total-asset fixed effects forces the parameters of interest, γ^h, to be identified solely from comparing similar sized firms within the same industry. The vector $X_{i,t}$ includes a set of time-varying firm-level control variables, such as log of market equity (size) at the end of May and cumulative 1-year excess returns (momentum). We additionally include up to three lags of the outcome and shock variables.

We cluster standard errors at the firm-level, which allows for a completely unrestricted specification of the residual covariance matrix in the time-series dimension. This effectively addresses the issue of serial correlation in residuals arising in a local projection framework.

Identifying assumptions and threats to identification The instrumental variable exclusion restriction in a local projection setting differs slightly from the usual one due to the dynamic structure of the problem. Identification requires a contemporaneous and a lead-lag exclusion re-

[17] Similarly, LP-IVs are often used to study the effects of monetary policy on investment or asset prices (e.g., in Jordà, Schularick, and Taylor, 2020, Kroen, Liu, Mian, and Sufi, 2021, and Bauer and Swanson, 2023).

striction. The instrument must be uncorrelated with past and future shocks, at least after including control variables. The exclusion restriction requires that assignment of index membership is exogenous and that changes in BMI only affect firm outcomes through changes in CAPM β. In Section 5 we find no evidence that changes in BMI correlate with changes in risk, financial constraints, or governance at the firm. Importantly, Column (1) of Appendix Table A6 shows that changes in BMI do not correlate with firm statements about delaying investments. Failure of the exclusion restriction would introduce bias in the estimated treatment effects. The size and sign of the bias depend on the size and sign of the failure and the strength of the instrument.[18] The literature on index inclusion effects generally finds positive but modest direct effects of index inclusion on real outcomes (e.g., Kacperczyk et al., 2021). Therefore, if anything, we expect an upward bias in our estimates.

To ensure that our estimates are well-identified, we follow three steps. First, we include up to three lags of outcomes and shock in our regressions. Second, we saturate our LP-IV estimator with high-dimensional fixed effects to remove as much time-varying unobserved heterogeneity as possible. Third, we verify that including a set of known predictors of capital accumulation (e.g., Tobin's Q or cash flow) does not change our results in a robustness test.

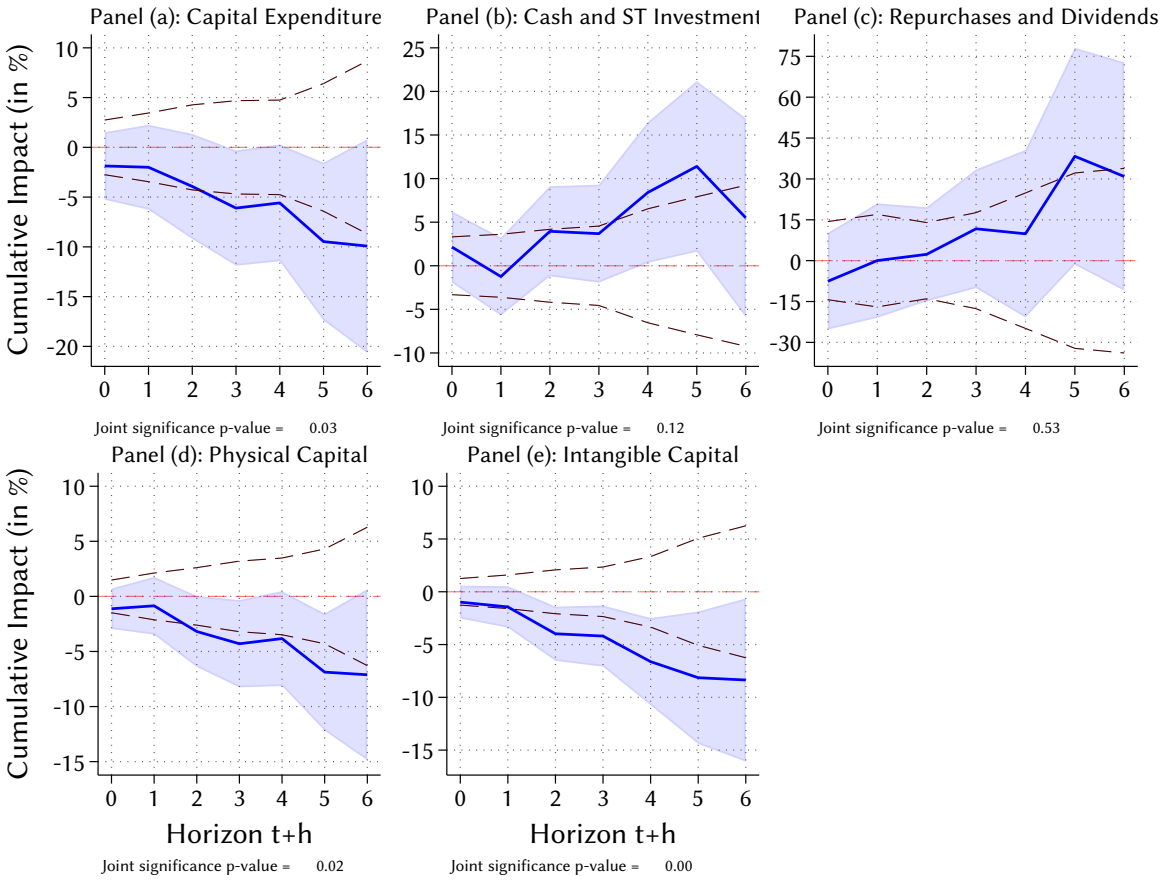
6.2 Results

Figure 3 shows estimates of the dynamic impulse response path of Eq. (12) to an increase in the CAPM β for capital expenditure, physical capital, intangible capital, and cash and short-term investments. The impulse is scaled to the average effect size for an increase of 5 p.p. in BMI in the Compustat sample (approximately a 15% increase in CAPM β^E). Note that the coefficients derived from local projection (LP) impulse responses are highly correlated. Similar to the case of regressions affected by near-collinearity, individual t-statistics may not appear significant, but a joint test for significance overwhelmingly rejects the null of zero treatment effect (see Jordà, 2023, for details). We therefore report F-statistics of joint-significant test and additionally display approximate 90% significance bands. The significance bands are calculated by inverting the statistic of the null that all impulse response coefficients are zero.

Several noteworthy facts are visible in Figure 3. First, the impulse responses across all outcome variables have the expected signs. Capital expenditure and physical and intangible capital

[18] Consider the example given by Jordà et al. (2020): Let y be the outcome, Δr the intervention, and z the instrument. The IV setup consists of the first and second stage given by $\Delta r = zb + \eta$ and $y = \widehat{\Delta r}\beta + z\phi + \nu$ where $\mathbb{E}[\Delta r \nu] \neq 0$ but $\mathbb{E}[z\nu] \neq 0$. The exclusion restriction assumes that $\phi = 0$. If $\phi \neq 0$, we have $\hat{\beta}_{IV} \xrightarrow{p} \beta + \phi/b$. The bias induced by failure of the exclusion restriction depends on both the size of the failure, ϕ, and the strength of the instrument, b. Weaker instruments tend to worsen the bias (see also Conley, Hansen, and Rossi, 2012).

Figure 3: LP-IV: Impulse Response of Outcome Variables to CAPM β shock



Notes: Figure shows LP-IV coefficient estimates for $100\times$ cumulative log-changes of outcome variables. Intervention is a shock to CAPM β estimated with change in BMI as an instrumental variable. Dashed red lines represent 90% significance bands computed by inverting the F-statistic of joint significance around zero using Scheffé's method.

stocks react negatively in response to an equity cost shock. Cash and short-term investments increase in response to the shock. Firms increase dividends and stock repurchases after several years. Second, the dynamic paths of firm responses to an increase in CAPM β are gradual and slow-moving. The responses are effectively zero in the treatment year $h = 0$ and then grow as time passes. The cumulative impact of the shock becomes statistically and economically significant after approximately 3 years. This aligns with the industry practice of using an equal-weighted rolling window of two to five years to estimate CAPM βs. Because older data points are included, the estimated cost of capital that managers use for investment decisions will gradually reflect changes in BMI, despite the "true" CAPM β of a firm's stock responding immediately. Third, the effects of an increase in CAPM β on capital expenditure and capital stocks are

Table 4: LP-IV: Point Estimates of the Real Effects of Changes in β

Horizon	Capital Expenditure	Cash & Short-Term Inv.	Dividends & Repurchases	Physical Capital	Intangible Capital	No Obs.
t+1	-2.00	-1.24	0.04	-0.85	-1.44	13,829
	(2.59)	(2.71)	(12.75)	(1.59)	(1.19)	
t+2	-3.93	3.96	2.33	-3.18	-3.98*	11,943
	(3.21)	(3.14)	(10.49)	(1.96)	(1.56)	
t+3	-6.10+	3.68	11.70	-4.30+	-4.20*	10,342
	(3.52)	(3.42)	(13.23)	(2.41)	(1.76)	
t+4	-5.58	8.41+	9.86	-3.83	-6.62**	8,881
	(3.57)	(4.91)	(18.66)	(2.61)	(2.51)	
t+5	-9.47*	11.40+	38.30	-6.87*	-8.14*	7,518
	(4.82)	(5.94)	(24.16)	(3.23)	(3.81)	
t+6	-9.92	5.50	30.89	-7.12	-8.36+	6,327
	(6.52)	(6.93)	(25.43)	(4.72)	(4.69)	
H_0: ATE = 0	0.03	0.12	0.53	0.02	0.00	
Avg. FS F-stat.			50.48			

Notes: This table reports coefficient estimates of LP-IV specifications of the form: $\log(Y_{i,t+h}) - \log(Y_{i,t-1}) = \alpha_i^h + \alpha_{j,t}^h + \gamma^h \Delta\widehat{\text{CAPM}}\,\beta_{i,t} + \xi^h X_{i,t} + \varepsilon_{i,t+h}$ for $h = 1, \ldots, 6$. Specifications include lags of outcome variables and shocks of up to 3th order. Variables are winsorized at the 2.5% and 97.5% level to remove outliers. FS F-stat refers to the Kleibergen-Paap F-stat for weak instruments. H_0 ATE $= 0$ refers to the null that the coefficients for $h = 1, \ldots, 6$ are jointly zero. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

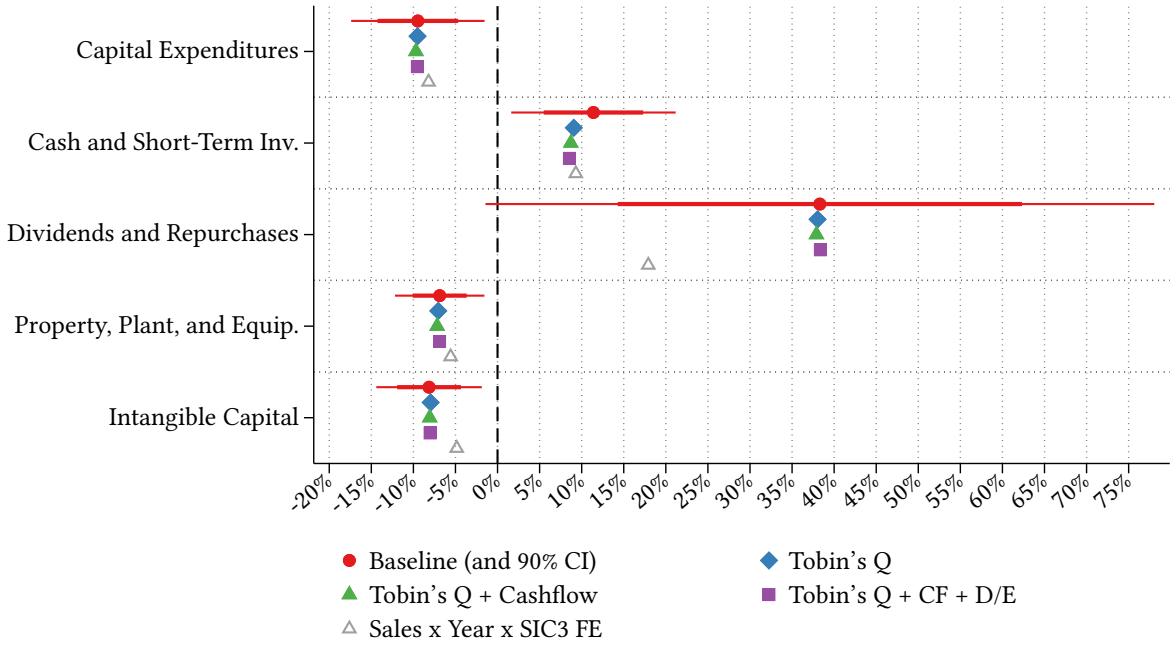
persistent and remain significant for at least six years after the shock.

Benchmarking-induced increases in CAPM βs lead to a large and persistent declines in investment. We find that firms reduce their capital expenditure by approximately 10.0% over six years in response to a shock to their CAPM β of 15%. The resulting decrease in physical capital stocks is 7.1% and in intangible capital stocks is 8.4% after six years. Point estimates after six years imply a user cost of capital elasticity (Zwick and Mahon, 2017) of around -1.09[19] and -1.28 for physical and intangible capital, respectively. These elasticities are close to the theoretical elasticity of 1 implied by a Cobb-Douglas production function.

Table 4 reports the estimates of the LP-IV regressions of Eq. (12) for the following outcome variables: (1) capital expenditure, (2) cash and short-term investments, (3) dividends and stock repurchases, (4) physical capital stocks, and (5) intangible capital stocks. We check whether the BMI instrument is weak with Kleibergen-Paap tests. We find that changes in BMI are strong

[19] Calculated as $-1.09 \approx \frac{-7.12\%}{100\% \times \frac{\left(\frac{0.16}{1+(1-0.30)\times.90} \times 600bps\right)}{904bps}}$, assuming a constant average tax rate of 30% (median in sample tax rate), constant debt-to-equity ratio equal to the median over time of 0.90 (median in sample debt-to-equity ratio), and an average cost of capital of 9.04% (median in sample perceived cost of capital adjusted for CAPM β distortion) and a ERP of 6% p.a..

Figure 4: LP-IV Robustness to various Alternative Specifications



Notes: Figure shows LP-IV coefficient estimates for $h = 5$ years of Eq. (12) for baseline and alternative specifications.

instruments for changes in CAPM β with average first-stage F-statistics above 50. Finally, we test the null hypothesis that all ATE coefficients are jointly zero, reporting the p-value of the test in the row labeled H_0: ATE $= 0$. All average treatment effects are statistically significant at conventional levels, except for dividends and repurchases; reflecting the noise also present in Figure 3.

Robustness Checks We perform several robustness checks on the main results. First, we add firm-level controls, such as cash flow, Tobin's Q, and the debt-to-equity ratio, which are known predictors of investment, to the LP-IV regressions. Second, we incorporate different levels of fixed effects, replacing firm size by industry by year fixed effects with sales by industry by year fixed effects. Figure 4 presents the coefficient estimates of these robustness checks at horizon $h = 5$ alongside the original estimate. The figure shows that adding firm-level controls does not change the point estimates, supporting our identification strategy. Altering the level of fixed effects only marginally affects the point estimates, with all changes well within one standard error of the original estimates.

Investment rates Appendix Figure A2 shows the impact of a 15% increase in CAPM β on firm investment rates[20]. The investment rate declines over time in response to the shock. Starting from a near-zero impact at $t+0$, gradually becoming more negative, reaching the lowest point after four to five years. The shock to the CAPM β leads to a significant (p<0.005) drop of -2.56 p.p. in investment rates after four years. In standard deviation units, firm investments drops by 0.19 standard deviations. In terms of economic magnitudes this compares to Alfaro, Bloom, and Lin (2024) who find that uncertainty shocks lead to a 0.18 standard deviation drop in investment rates.

6.3 Changes in CAPM β and manager's perceived cost of capital

We document that changes in the CAPM β affect the perceived cost of capital of firm managers. We use a firm's self-reported perceived cost of capital collected from earnings calls by Gormsen and Huber (2024). The authors find substantial excess volatility in the perceived cost of equity. We argue that this excess volatility is partly driven by changes in the firm's CAPM β induced by high BMI and is orthogonal to aggregate risk.

We estimate the pass-through of changes in a firm's CAPM β on the perceived cost of capital using a series of instrumental variable (IV) regressions of the following form:

$$\Delta\text{CAPM } \beta_{i,t} = \delta_i + \delta_{j,t} + \theta \Delta\text{BMI}_{i,t} + \epsilon_{i,t} \tag{13}$$

$$\Delta\text{Perceived Cost of Capital}_{i,t} = \alpha_i + \alpha_{j,t} + \sum_{h=0}^{4} \gamma_h \Delta\widehat{\text{CAPM}}\,\beta_{i,t-h} + \varepsilon_{i,t} \tag{14}$$

where the coefficient of interest, $\sum_{h=0}^{4}\gamma_h$, provides a causal estimate of the pass-through of changes in a firm's CAPM β to its perceived cost of capital. Including firm fixed effects α_i ensures we remove time-invariant heterogeneity across firms, and, in particular, accounts for possible ex-ante differences in characteristics between treated and control firms. Size-by-year, liquidity-by-year, or industry-by-year fixed effects, $\alpha_{j,t}$, further restrict the identifying variation to comparing firms within the same size decile, liquidity decile, or industry each period.

We use a distributed lag model in Eq. (14) for both economic and econometric reasons. As argued above, managers typically estimate CAPM β using equal-weighted OLS regressions over two to five-year horizons. Changes in CAPM β are therefore not immediately reflected in managers' perceived cost of capital but gradually incorporated as new data becomes available. Gormsen and Huber (2023) show that managers' perceived cost of capital affects their required returns on new investments (i.e., their discount rates) and ultimately their investment decisions. This transmis-

[20] Defined as $\frac{\text{CAPX}_t}{\frac{1}{2}(\text{PPENT}_{t-1}+\text{PPENT}_t)}$ (see e.g, Belo, Lin, and Bazdresch, 2014).

Table 5: Effect of ΔCAPM β on Managers' Perceived Cost of Capital (Gormsen and Huber, 2024)

	RF			OLS			IV		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Dependent variable: Δ Perceived Cost of Capital								
Δ BMI	1.643***	1.680***	1.667***						
	(0.375)	(0.373)	(0.402)						
Δ CAPM β^A				1.202***	1.183***	1.135***	3.153***	3.694***	3.148***
				(0.057)	(0.059)	(0.075)	(0.825)	(0.920)	(0.769)
Firm	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time	✓			✓			✓		
Size Quartile × Time		✓			✓			✓	
Industry × Time			✓			✓			✓
Observations	19,501	19,501	18,940	19,501	19,501	18,940	19,501	19,501	18,940

Notes: This table reports regression estimates for specifications of the form: $\Delta\text{Perc. Cost of Capital}_{i,t} = \alpha_i + \alpha_{j,t} + \sum_{h=0}^{4} \gamma_h \Delta\text{CAPM } \beta_{i,t-h} + \varepsilon_{i,t}$ for reduced form, ordinary least squares, and instrumental variable regression where the instrument is ΔBMI for stock i in year t. IV estimated via LIML. Yearly sample from 2000 to 2018. Standard errors in parentheses are clustered at firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

sion can be muted in the short run, as shocks to perceived cost of capital are transmitted into firm's discount rates with several years of lag. In addition, the data strongly favors a distributed lag model over a contemporaneous effect model. Appendix Table A8 presents Bayesian information criterion (BIC) values for lag orders from h=0 to h=6, demonstrating that the data supports including fourth-order lags.

A firm's perceived cost of capital and its stock's CAPM β are jointly determined by its exposure to aggregate risk in equilibrium. We therefore employ an instrumental variable approach to address this endogeneity and potential attenuation bias due to measurement error. The change in BMI is a natural instrument in this setting because it induces change in the CAPM β that is orthogonal to aggregate market risk. The results of the difference-in-differences from above establish that changes in BMI are relevant instruments for changes in a firm's CAPM β.

The exclusion restriction requires that an increase in BMI affects a firm's perceived cost of capital only through an increase in the firm's CAPM β. A violation of the exclusion restriction could occur if BMI affects a manager's perceived cost of capital through channels other than changes in the firm's CAPM β. For example, an increase in BMI could lower a firms' perceived cost of capital through a reduction in credit spreads because of reputational effects. In Section 5 we find no evidence that changes in BMI correlate with changes in risk, financial constraints, or governance at the firm. We interpret this as evidence that the exclusion restriction is not violated.

Table 5 reports the results of Eq. (14) for reduced form (RF), OLS, and IV specifications. Columns (1) to (3) report RF, columns (4) to (6), OLS estimates and columns (7) to (9) report

IV estimates. The coefficient estimates are positive, stable across specifications, and always statistically significant at the 0.1% level. Columns (1) to (3) of Table 5 show that increases in BMI predict increases in managers' perceived cost of capital.

Comparing the OLS estimates to the IV estimates in Table 5 shows that OLS estimates are downward-biased. Estimates using changes in BMI as an instrument for changes in CAPM β are approximately three times the magnitude of OLS estimates. Classical measurement error likely explains the downward bias. We propose two mechanisms that introduce measurement error. First, we do not observe how firms estimate their CAPM β when calculating their perceived cost of capital. We use the CAPM β estimates of Welch (2022b) due to their superior performance in predicting future CAPM βs out of sample. In practice, firms often simply estimate CAPM βs using equally-weighted observations on a rolling window of two to five years (Berk and DeMarzo, 2023). Second, any empirical measure of CAPM β is an *estimate* of the true CAPM β and thus subject to measurement error.[21]

The IV estimates in column (8) imply that a two standard deviation change (≈ 0.20) in a firm's CAPM β increases the manager's perceived cost of capital by 80 bps. This is similar to the change in CAPM β observed after a 10 to 20 p.p. increase in BMI following index inclusion (see Figure 2). The resulting increase in perceived cost of capital is substantial compared to typical financial price fluctuations. For example, Krishnamurthy and Vissing-Jorgensen (2011) find that the Fed's quantitative easing reduced corporate bond yields by up to 50 bps.

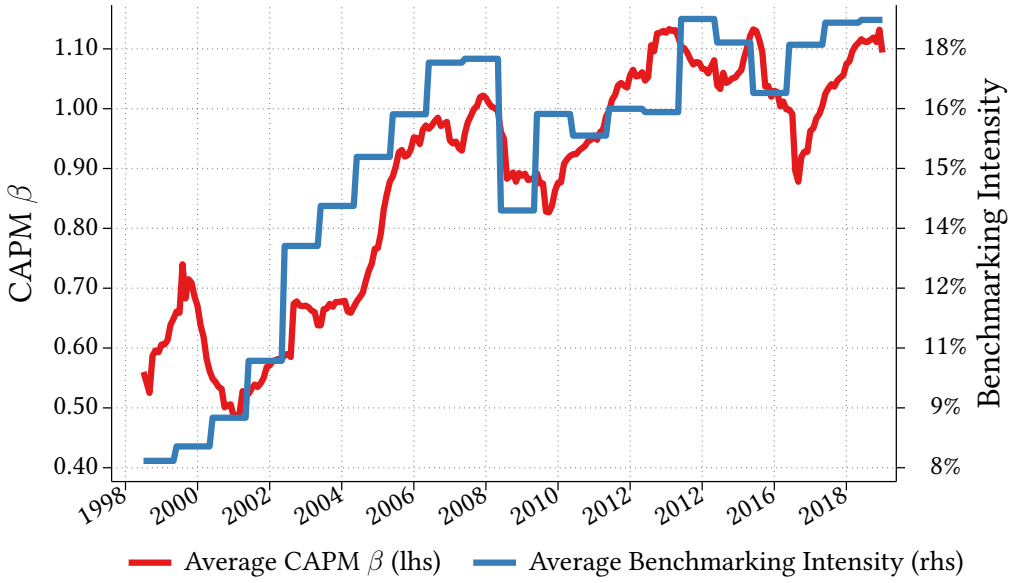
7 Aggregate Effects of increased Benchmarking

A body of literature finds that U.S. investment has been low in recent decades relative to profitability and valuations. Specifically, the secular decline in risk-free interest rates has increased firms' market value and raised Tobin's Q since the early 2000s. According to standard Q-theory, investment should have increased with the rise in Tobin's Q (Gutiérrez and Philippon, 2017). However, investment rates have been low by historical standards, leading researchers to argue that there is "missing investment", now exceeding 20% of the aggregate capital stock, even after accounting for intangibles and other measurement issues (Gormsen and Huber, 2023).

We argue that the increase in BMI has contributed to the missing investment puzzle. Our difference-in-differences results show that increases in BMI due to index inclusion increase the CAPM βs of firms. Extending our scope to the whole cross-section of firms, we estimate that

[21] An alternative way to address this measurement error is to use estimators which rely on higher cumulants, as proposed by Erickson, Jiang, and Whited (2014). We opt to use IV regressions because of their simplicity.

Figure 5: Average CAPM β and BMI over time



(a) Aggregate increase in CAPM βs due to increasing BMI

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$\overline{\text{BMI}}_t$ (%)	$\overline{\text{CAPM } \beta}_t$	Cons.	Adj. R^2	$\overline{\text{BMI}}_{1998}$	$\overline{\text{BMI}}_{2018}$	$\Delta\ \overline{\text{BMI}}_{'98\to'18}$	$\Delta\widehat{\overline{\text{CAPM } \beta}}_{'98\to'18}$
$\overline{\text{CAPM } \beta}_{t+1}$	0.023***	0.538***	0.059	0.86	8.16%	18.32%	10.16 p.p.	0.231***
	(0.004)	(0.112)	(0.059)		(0.10)	(0.16)		(0.042)

Notes: Columns (1), (2), (3) report estimates for γ_1, γ_2, and α of the following regression: CAPM $\beta_{t+1} = \alpha + \gamma_1$ BMI$_t + \gamma_2$ CAPM $\beta_t + \varepsilon_{t+1}$. Newey-West standard error in parentheses with L=2 lags. Yearly sample from 1998 to 2018 (N=21). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001. Column (5) and (6) report the average BMI in 1998 and 2018, respectively. Column (7) is the increase in BMI over the sample period. Column (8) reports the predicted increase in the CAPM β due to increasing BMI over the sample period.

the average CAPM β of firms has increased by 0.46 over the past 20 years. Using time-series techniques, we attribute 50% of this increase to the increase in average BMI from 8.2% in 1998 to 18.3% in 2018.

We use the NBER-CES Manufacturing Industry Database to estimate the effect of increases in industries' average CAPM β on capital accumulation. Results that are weighted by value-added indicate that higher CAPM βs led to 7.6% lower capital accumulation from 2000 to 2016 at the aggregate-level. Increases in CAPM βs can thus explain approximately 43% of the missing investment.

7.1 Increase in average CAPM β over the past 20 years

Figure 5 shows the evolution of average CAPM β and average BMI from 1998 to 2018. A casual inspection of the graphs suggests that both time-series are related. The average BMI increased from 8.16% in 1998 to 18.32% in 2018, whereas the average CAPM β increased from 0.63 to 1.09 over the same time window. Indeed, our main results establishes that increases in BMI at the stock-level cause the CAPM β to increases.

How much of the increase in average CAPM β is due to the increase in average BMI? We estimate how increases in the average cross-sectional BMI affect the average cross-sectional CAPM β using the following regression specification:

$$\overline{\text{CAPM } \beta}_{t+1} = \alpha + \gamma_1 \, \overline{\text{BMI}}_t + \gamma_2 \, \overline{\text{CAPM}\beta}_t + \varepsilon_t. \tag{15}$$

To address the potential of a "spurious regression", we follow the advice of Hansen (p. 588, 2023) and control for the lagged value of the outcome variable.

Table 5a reports results of this time-series regressions in Columns (1) to (4). Column (1) shows that BMI is a statistically significant (p<0.001) predictor of CAPM β, even after considering the high autocorrelation of CAPM βs. A 1 p.p. increase in average BMI is associated with a 0.023 increase in the average CAPM β. Column (7) shows that the average BMI has increased by 10.16 p.p. from 1998 to 2018. Column (8) shows that a 10.16 p.p. increase in average BMI is associated with an 0.231 increase in average CAPM β. This implies that approximately 50% of the increase in average CAPM β from 1998 to 2018 is driven by an increase in average BMI. An increase in the average CAPM β of 0.231 increase further implies that firms, on average, perceive their equity cost of capital to have increased by 139 bps (assuming an ERP of 6%). The time-series estimate of a 0.231 is nearly identical to the causally identified difference-in-differences estimate of 0.23 for a similar size increase in BMI, as shown in Figure 2.

7.2 Long-term effects of CAPM β distortions on capital accumulation

We use the NBER CES Manufacturing Industry database to study the long-term effects of increasing CAPM βs due to increased benchmarking on capital accumulation at the aggregate-level.

We estimate the long-term effects of increasing CAPM βs due to higher benchmarking at the NAICS-5 digit industry-level. To obtain results for the aggregate economy, we weigh each industry observation by its value-added in the year 2000. We estimate a series of IV regressions

in long-difference from 2000 to 2016 of the form:

$$\Delta\text{CAPM } \beta_i^{`00 \to `16} = \delta_j + \theta \Delta \text{BMI}_i^{`00 \to `16} + \zeta X_i^{`00} + \epsilon_i \quad (16)$$

$$\log\left(\text{Real Capital Stock}_i^{`16} / \text{Real Capital Stock}_i^{`00}\right) = \alpha_j + \gamma \Delta \widehat{\text{CAPM}}\, \beta_i^{`00 \to `16} + \zeta X_i^{`00} + \varepsilon_i \quad (17)$$

where α_j are NAICS 3-digit subsector fixed effects. The vector of control variables $X_i^{`00}$ includes industry-level characteristics in 2000, such as the log of employment and TFP. The controls account for initial differences in industry characteristics that may affect capital accumulation. We calculate the change in CAPM β for industry i from 2000 to 2016 as the difference between the market equity-weighted average CAPM β of firms in that industry in 2016 and 2000. Similarly, we calculate the change in industry BMI from 2000 to 2016 as the difference between the market equity-weighted average BMI of firms in the same industry in 2016 and 2000.

A potential concern with our industry-level analysis is that the effects of increasing benchmarking on capital accumulation may be confounded by other secular changes in the economy. Perhaps the most significant change over the sample period was the U.S. economy's transition from a manufacturing-based economy to a technology- and information-based economy.

We address this concern by including NAICS 3-digit sub-sector[22] fixed effects in the regressions. Using sub-sector fixed effects, we identify the effects of changes in CAPM β on capital accumulation using variation across industries within the same sub-sectors.

Results Table 6 reports coefficient estimates of Eq. (17) for OLS and WLS, with observations weighted by their value added in 2000. Several things are worth noting. First, across all specifications, we find that increases in CAPM βs have a statistically significant (p<0.05) negative effect on long-term capital accumulation at the industry-level. Second, the estimated coefficients are economically meaningful. The weighted coefficient of Column (7) implies that a 0.23 increase in average CAPM β is associated with a 7.6% ($\approx 0.23 \times 0.33 \times 100\%$) lower aggregate capital stock over the 17-year period. These magnitudes imply a user cost of capital elasticity of approximately -1.04,[23] which is close to the value of 1 implied by a Cobb-Douglas production function. Third, the results are robust to the inclusion of industry-level controls and sub-sector fixed effects. Fourth, changes in BMI are strong instruments for changes in CAPM β even at the industry-level, with first-stage F-stats of at least 17.7.

[22] For example, the NAICS 3-digit sub-sector "311 - Food Manufacturing" contains 12 NAICS 5-digit industries.

[23] Calculated as $-1.04 \approx \frac{-7.6\%}{100\% \times \frac{\left(\frac{0.23}{1+(1-0.35)\times 1.4} \times 600bps\right)}{1000bps}}$, assuming a constant average tax rate of 35%, constant debt-to-equity ratio equal to the median over time of 1.40, and an average cost of capital of 10% and a ERP of 6% p.a..

Table 6: Long-term Effects of Benchmarking on Capital Accumulation at the Industry-level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: $\log(\text{Real Capital Stock}^{`16})$				$-\log(\text{Real Capital Stock}^{`00})$			
ΔCAPM $\beta^{`00\to`16}$	-0.327^{+}	-0.300^{+}	-0.484*	-0.525**	-0.280*	-0.276*	-0.322*	-0.293*
	(0.172)	(0.155)	(0.188)	(0.195)	(0.132)	(0.128)	(0.147)	(0.147)
$\log(\text{Emp}^{`00})$		-0.041		-0.003		0.0054		0.072*
		(0.034)		(0.037)		(0.035)		(0.030)
$\log(\text{TFP}^{`00})$		0.057		-0.341^{+}		0.383**		-0.019
		(0.150)		(0.175)		(0.134)		(0.116)
Constant	0.266*	0.428*			0.213***	0.193		
	(0.104)	(0.211)			(0.063)	(0.213)		
Fixed Effects								
Subsector			✓	✓			✓	✓
FS F-stat.	20.14	26.33	17.71	18.32	25.64	30.70	29.23	27.29
Weights					VA$^{`00}$	VA$^{`00}$	VA$^{`00}$	VA$^{`00}$
Observations	111	111	107	107	111	111	107	107

Notes: This table reports coefficient estimates of regression at the NAICS 5-digit industry-level of the form: $\Delta \log(\text{Real Capital Stock})_i^{`00\to`16} = \alpha_j + \gamma \Delta\text{CAPM}\ \beta_i^{`00\to`16} + \zeta X_i^{`00} + \varepsilon_i$ where changes in CAPM β are instrumented with changes in BMI. Sub-sector fixed effects are at the NAICS 3-digit level. Columns (5) to (8) weighted by industry value added in 2000. FS F-stat refers to the Kleibergen-Paap F-stat for weak instruments. All variables are winsorized 1% and 99% level to account for outliers. Standard errors in parentheses. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Gormsen and Huber (Figure 10, 2023) calculate that the cumulative investment shortfall in the U.S. from 2002 to 2016 is approximately 17.5%. Our results suggest that the increase in CAPM βs due to the increase in benchmarking can explain approximately 43% of the cumulative investment shortfall.

8 Conclusion

In this paper, we study the real effects benchmarking-induced asset price distortions on firm investment. We provide causal evidence that increases in benchmarking intensity cause increases in CAPM β for individual stocks, and in the aggregate. Our estimates imply that increases in aggregate benchmarking intensity caused the average CAPM β to increase by 0.23 over the past 20 years. A battery of robustness checks confirms that our findings are not driven by changes in firm or industry fundamentals.

Using an instrumental variable approach, we find that firms respond to benchmarking-induced increases in CAPM β by reducing investment. We argue that these findings are driven by firm managers' failure to internalize benchmarking-induced asset price distortions. Instead, man-

agers rely on textbook guidance to estimate costs of capital using the CAPM. Consistent with this mechanism, firm's perceived costs of capital reported in earnings calls are responsive to benchmarking-induced increases in CAPM β.

A body of literature shows that U.S. investment has been low relative to valuations in the past two decades. Increases in average CAPM β can explain approximately 43% of the cumulative investment shortfall since the early 2000s. Our findings show that the benchmarking of asset managers has large and economically meaningful implications for aggregate investment.

References

Aghion, P., J. Van Reenen, and L. Zingales (2013): "Innovation and Institutional Ownership," *American Economic Review*, 103, 277–304.

Alfaro, I., N. Bloom, and X. Lin (2024): "The Finance Uncertainty Multiplier," *Journal of Political Economy*, 132, 577–615.

Andrei, D., J. Cujean, and M. Wilson (2023): "The Lost Capital Asset Pricing Model," *Review of Economic Studies*, 90, 2703–2762.

Antón, M. and C. Polk (2014): "Connected Stocks," *Journal of Finance*, 69, 197–226.

Appel, I. R., T. A. Gormley, and D. B. Keim (2016): "Passive Investors, Not Passive Owners," *Journal of Financial Economics*, 121, 111–141.

Asker, J., J. Farre-Mensa, and A. Ljungqvist (2014): "Corporate Investment and Stock Market Listing: A Puzzle?" *The Review of Financial Studies*, 28, 342–390.

Azar, J., M. C. Schmalz, and I. Tecu (2018): "Anticompetitive Effects Of Common Ownership," *The Journal of Finance*, 73, 1513–1565.

Baker, A. C., D. F. Larcker, and C. C. Wang (2022): "How Much Should We Trust Staggered Difference-in-differences Estimates?" *Journal of Financial Economics*, 144, 370–395.

Bansal, R., S. Miller, D. Song, and A. Yaron (2021): "The Term Structure Of Equity Risk Premia," *Journal of Financial Economics*, 142, 1209–1228.

Barberis, N., A. Shleifer, and J. Wurgler (2005): "Comovement," *Journal of Financial Economics*, 75, 283–317.

Basak, S. and A. Pavlova (2013): "Asset Prices and Institutional Investors," *American Economic Review*, 103, 1728–58.

Bauer, M. D. and G. D. Rudebusch (2020): "Interest Rates Under Falling Stars," *American Economic Review*, 110, 1316–1354.

Bauer, M. D. and E. T. Swanson (2023): "A Reassessment of Monetary Policy Surprises and High-Frequency Identification," *NBER Macroeconomics Annual*, 37, 87–155.

Bebchuk, L. A., A. Cohen, and S. Hirst (2017): "The Agency Problems Of Institutional Investors," *Journal of Economic Perspectives*, 31, 89–102.

Bebchuk, L. A. and S. Hirst (2019): "Index Funds and the Future Of Corporate Governance: Theory, Evidence, and Policy," NBER Working Paper 26543 26543, National Bureau of Economic Research.

Belo, F., X. Lin, and S. Bazdresch (2014): "Labor Hiring, Investment, and Stock Return Predictability In the Cross Section," *Journal of Political Economy*, 122, 129–177.

Ben-David, I. and A. M. Chinco (2023): "Modeling Managers As EPS Maximizers," NBER Working Paper 31125, National Bureau of Economic Research.

Ben-David, I., F. Franzoni, and R. Moussawi (2018): "Do ETFs Increase Volatility?" *The Journal of Finance*, 73, 2471–2535.

Beneish, M. D. and R. E. Whaley (1996): "An Anatomy Of the "S&P Game": the Effects Of Changing the Rules," *Journal of Finance*, 51, 1909–30.

Berk, J. B. and P. M. DeMarzo (2023): *Corporate Finance*, Boston: Pearson Education, 6th ed.

Berk, J. B. and J. H. Van Binsbergen (2016): "Assessing Asset Pricing Models Using Revealed Preference," *Journal of Financial Economics*, 119, 1–23.

Blume, M. E. (1975): "Betas and Their Regression Tendencies," *The Journal of Finance*, 30, 785–795.

Brealey, R., S. Myers, F. Allen, and A. Edmans (2023): *Principles Of Corporate Finance. 14*, New York: McGraw Hill, 14th ed.

Buffa, A. M., D. Vayanos, and P. Woolley (2022): "Asset Management Contracts and Equilibrium Prices," *Journal of Political Economy*, 130, 3146–3201.

Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019): "The Effect Of Minimum Wages on Low-wage Jobs," *The Quarterly Journal of Economics*, 134, 1405–1454.

Chang, Y.-C., H. Hong, and I. Liskovich (2014): "Regression Discontinuity and the Price Effects of Stock Market Indexing," *The Review of Financial Studies*, 28, 212–246.

——— (2015): "Regression Discontinuity and the Price Effects Of Stock Market Indexing," *The Review of Financial Studies*, 28, 212–246.

Conley, T. G., C. B. Hansen, and P. E. Rossi (2012): "Plausibly Exogenous," *Review of Economics and Statistics*, 94, 260–272.

De Chaisemartin, C. and X. d'Haultfoeuille (2023): "Two-way Fixed Effects and Differences-

in-differences with Heterogeneous Treatment Effects: a Survey," *The Econometrics Journal*, 26, C1–C30.

Deshpande, M. and Y. Li (2019): "Who Is Screened Out? Application Costs and the Targeting Of Disability Programs," *American Economic Journal: Economic Policy*, 11, 213–248.

Dessaint, O., J. Olivier, C. A. Otto, and D. Thesmar (2020): "CAPM-Based Company (Mis)valuations," *The Review of Financial Studies*, 34, 1–66.

Dimson, E. (1979): "Risk Measurement When Shares Are Subject to Infrequent Trading," *Journal of Financial Economics*, 7, 197–226.

Edmans, A. and C. G. Holderness (2017): "Blockholders: A survey of theory and evidence," in *The Handbook of the Economics of Corporate Governance*, ed. by B. E. Hermalin and M. S. Weisbach, Elsevier, vol. 1, chap. 8, 541–636.

Erickson, T., C. H. Jiang, and T. M. Whited (2014): "Minimum Distance Estimation Of the Errors-in-variables Model Using Linear Cumulant Equations," *Journal of Econometrics*, 183, 211–221.

Freyaldenhoven, S., C. Hansen, J. P. Pérez, and J. M. Shapiro (2021): "Visualization, Identification, and Estimation In the Linear Panel Event-study Design," NBER Working Paper 29170, National Bureau of Economic Research.

Giglio, S., B. T. Kelly, and S. Kozak (2023): "Equity Term Structures Without Dividend Strips Data," NBER Working Paper 31119, National Bureau of Economic Research.

Gomes, J., L. Kogan, and L. Zhang (2003): "Equilibrium Cross Section of Returns," *Journal of Political Economy*, 111, 693–732.

Gormley, T. A. and D. A. Matsa (2011): "Growing Out Of Trouble? Corporate Responses to Liability Risk," *The Review of Financial Studies*, 24, 2781–2821.

Gormsen, N. J. and K. Huber (2023): "Corporate Discount Rates," NBER Working Paper 31329, National Bureau of Economic Research.

——— (2024): "Firms' Perceived Cost Of Capital," NBER Working Paper 32611, National Bureau of Economic Research.

Graham, J. R. (2022): "Presidential Address: Corporate Finance and Reality," *The Journal of Finance*, 77, 1975–2049.

Grossman, S. J. and J. E. Stiglitz (1980): "On the Impossibility Of Informationally Efficient Markets," *The American Economic Review*, 70, 393–408.

Gutiérrez, G. and T. Philippon (2017): "Investmentless Growth: an Empirical Investigation," *Brookings Papers on Economic Activity*, Fall, 89–169.

Hansen, B. E. (2023): *Econometrics*, Princeton: Princeton University Press, 1st ed.

Harris, L. and E. Gurel (1986): "Price and Volume Effects Associated with Changes in the S&P 500 List: New Evidence for the Existence Of Price Pressures," *The Journal of Finance*, 41, 815–829.

Hassan, T. A., S. Hollander, L. Van Lent, and A. Tahoun (2019): "Firm-level Political Risk: Measurement and Effects," *The Quarterly Journal of Economics*, 134, 2135–2202.

Heath, D., D. Macciocchi, R. Michaely, and M. C. Ringgenberg (2021): "Do Index Funds Monitor?" *The Review of Financial Studies*, 35, 91–131.

Hoberg, G. and V. Maksimovic (2015): "Redefining Financial Constraints: a Text-based Analysis," *The Review of Financial Studies*, 28, 1312–1352.

Jacobs, M. T. and A. Shivdasani (2012): "Do You Know Your Cost Of Capital?" *Harvard Business Review*, 90, 118–124.

Jordà, Ò. (2005): "Estimation and Inference Of Impulse Responses by Local Projections," *American Economic Review*, 95, 161–182.

——— (2023): "Local Projections for Applied Economics," *Annual Review of Economics*, 15, 607–631.

Jordà, Ò., M. Schularick, and A. M. Taylor (2020): "The Effects Of Quasi-random Monetary Experiments," *Journal of Monetary Economics*, 112, 22–40.

Kacperczyk, M., S. Sundaresan, and T. Wang (2021): "Do Foreign Institutional Investors Improve Price Efficiency?" *The Review of Financial Studies*, 34, 1317–1367.

Karolyi, G. A. (1992): "Predicting Risk: Some New Generalizations," *Management Science*, 38, 57–74.

Kashyap, A. K., N. Kovrijnykh, J. Li, and A. Pavlova (2021): "The Benchmark Inclusion Subsidy," *Journal of Financial Economics*, 142, 756–774.

Krishnamurthy, A. and A. Vissing-Jorgensen (2011): "The Effects Of Quantitative Easing on

Interest Rates: Channels and Implications for Policy," *Brookings Papers on Economic Activity*, Fall, 215–287.

Kroen, T., E. Liu, A. R. Mian, and A. Sufi (2021): "Falling Rates and Rising Superstars," NBER Working Paper 29368, National Bureau of Economic Research.

Lewellen, J. and K. Lewellen (2022): "Institutional Investors and Corporate Governance: the Incentive to Be Engaged," *Journal of Finance*, 77, 213–264.

Limburg, A. (2024): "Do Firms Self-select Into Russell Indices?" Working paper, Available at SSRN 4816038.

Linn, M. and D. Weagley (2021): "Uncovering Financial Constraints," *Journal of Financial and Quantitative Analysis*, 1–36.

McCrary, J. (2008): "Manipulation Of the Running Variable In the Regression Discontinuity Design: a Density Test," *Journal of Econometrics*, 142, 698–714.

Modigliani, F. and M. H. Miller (1958): "The Cost Of Capital, Corporation Finance and the Theory Of Investment," *The American Economic Review*, 48, 261–297.

Pavlova, A. and T. Sikorskaya (2023): "Benchmarking Intensity," *The Review of Financial Studies*, 36, 859–903.

Peters, R. H. and L. A. Taylor (2017): "Intangible Capital and the Investment-Q Relation," *Journal of Financial Economics*, 123, 251–272.

Ross, S., R. Westerfield, J. Jaffe, and B. Jordan (2016): *Corporate Finance*, New York: McGraw Hill, 11th ed.

Shleifer, A. (1986): "Do Demand Curves for Stocks Slope Down?" *The Journal of Finance*, 41, 579–590.

Van Binsbergen, J., W. Hueskes, R. Koijen, and E. Vrugt (2013): "Equity Yields," *Journal of Financial Economics*, 110, 503–519.

Vasicek, O. A. (1973): "A Note on Using Cross-sectional Information In Bayesian Estimation Of Security Betas," *The Journal of Finance*, 28, 1233–1239.

Vijh, A. M. (1994): "S&P 500 Trading Strategies and Stock Betas," *The Review of Financial Studies*, 7, 215–251.

Welch, I. (2022a): *Corporate Finance*, Self-published, 5th ed.

——— (2022b): "Simply Better Market Betas," *Critical Finance Review*, 11, 37–64.

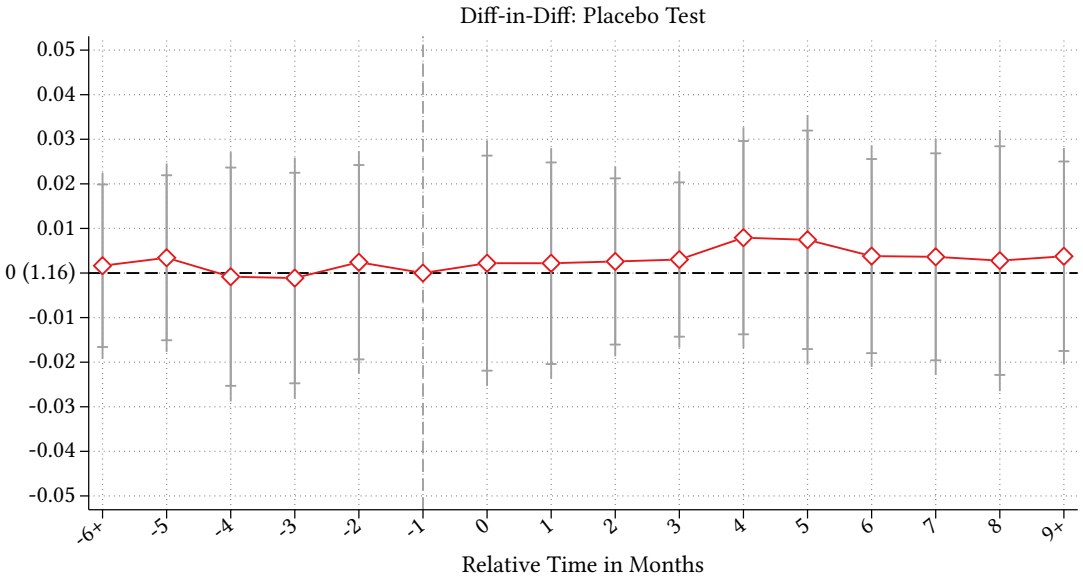
Wurgler, J. (2010): "on the Economic Consequences Of Index-linked Investing," NBER Working Paper 16376, National Bureau of Economic Research, Inc.

Zwick, E. and J. Mahon (2017): "Tax Policy and Heterogeneous Investment Behavior," *American Economic Review*, 107, 217–248.

A Appendix

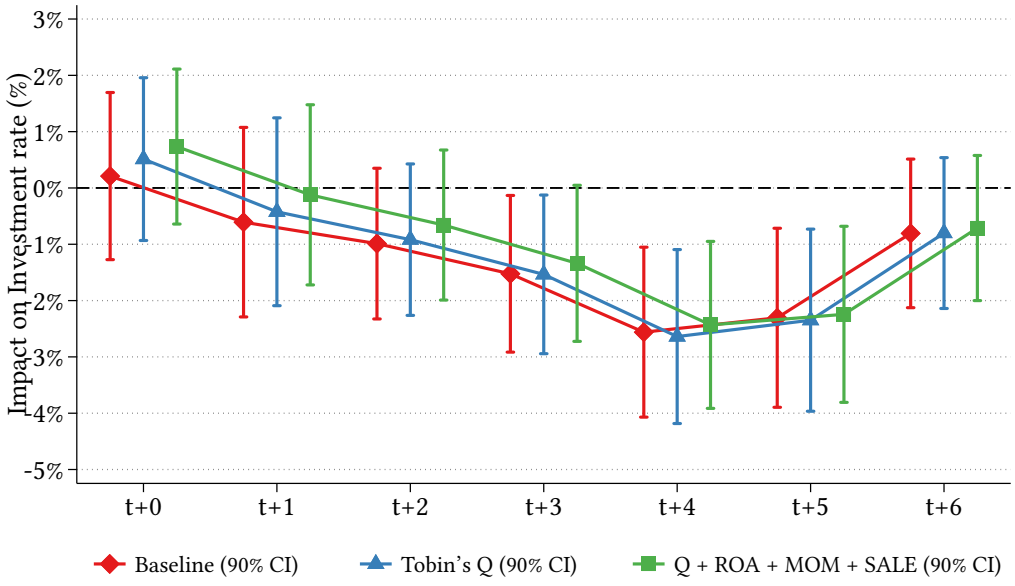
A.1 Appendix Figures

Figure A1: Event Study Placebo



Notes: Treatment group: ΔBMI $\in (0$ p.p., 1 p.p.$]$. Control group: ΔBMI $\in [-1$ p.p., 0 p.p.$]$

Figure A2: Impact of changes in CAPM β on investment rate



Notes: This figure shows γ^h coefficient estimates of Investment Rate$_{i,t+h} = \alpha_i + \alpha_{j,t} + \gamma^h \widehat{\text{CAPM}}\ \beta_{i,t} + X'_{i,t}\xi + \varepsilon_{i,t+h}$, estimated using changes in BMI as an instrumental variable for changes in CAPM β. Estimates are scaled to a 0.16 change in CAPM β. Investment rate is defined as $\frac{\text{CAPX}_t}{\frac{1}{2}(\text{PPENT}_{t-1}+\text{PPENT}_t)}$ (see e.g, Alfaro et al., 2024 or Belo et al., 2014). Median and standard deviation of investment rate is 19.9% and is 13.5% over the sample period from 2000 to 2018, respectively.

A.2 Appendix Tables

Table A1: Effect of Change in Benchmarking Intensity on CAPM β

Treatment Group:	Δ BMI>5p.p.							
Sample:	Market Equity $\geq$ P10 NYSE				Market Equity $\geq$ P20 NYSE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated $\times$ Post	0.111***	0.117***	0.111***	0.111***	0.102***	0.109***	0.102***	0.103***
	(0.013)	(0.012)	(0.012)	(0.012)	(0.014)	(0.012)	(0.013)	(0.013)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Fixed Effects								
Firm $\times$ Cohort	✓	✓	✓	✓	✓	✓		
Time $\times$ Cohort	✓				✓			
Size Decile $\times$ Time $\times$ Cohort		✓				✓		
Volume Decile $\times$ Time $\times$ Cohort			✓				✓	
Shrs. Out. Decile $\times$ Time $\times$ Cohort				✓				✓
Observations	291,822	291,754	291,688	291,617	249,916	249,840	249,745	249,703

Notes: Treated$\times$Post is the average of the post-treatment coefficients after 5 months (to account for the rolling-window estimation of CAPM β). Sample from 2000-01 to 2019-06. Standard errors in parentheses are double clustered at firm and year-month level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A2: Robustness to Alternative CAPM β Estimators

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Panel A: CAPM β estimates using expanding windows of daily data with exponentially decaying weights of 3-months half life						
	Δ CAPM β^{OLS}_{EW}	Δ CAPM β^{WEL}_{EW}	Δ CAPM β^{DIM}_{EW}	Δ CAPM β^{BLU}_{EW}	Δ CAPM β^{TOP}_{EW}	Δ $\rho(r_i, r_m)_{EW}$	Δ σ^i_{EW}
Δ BMI	1.146***	0.947***	0.752***	0.764***	1.056***	0.388***	-0.00222
	(0.059)	(0.047)	(0.080)	(0.039)	(0.055)	(0.019)	(0.002)
Firm FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.19	0.20	0.14	0.19	0.30	0.38	0.55
Observations	36,095	36,095	36,095	36,095	36,095	36,095	36,095
	Panel B: CAPM β estimates using daily data with a 2-year rolling window (with equal weights)						
	Δ CAPM β^{OLS}_{RW}	Δ CAPM β^{WEL}_{RW}	Δ CAPM β^{DIM}_{RW}	Δ CAPM β^{BLU}_{RW}	Δ CAPM β^{TOP}_{RW}	Δ $\rho(r_i, r_m)_{RW}$	Δ σ^i_{RW}
Δ BMI	0.300***	0.247***	0.212***	0.200***	0.325***	0.140***	-0.00429***
	(0.040)	(0.032)	(0.055)	(0.027)	(0.038)	(0.013)	(0.001)
Firm FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.24	0.25	0.18	0.24	0.33	0.39	0.53
Observations	36,095	36,095	36,095	36,095	36,095	36,095	36,095

Notes: This table reports coefficient estimates of specifications of the form: $\Delta\beta_{i,t} = \alpha_i + \alpha_t + \gamma\Delta\text{BMI}_{i,t} + \varepsilon_{i,t}$. β^{WEL} is estimator of Welch (2022b), β^{DIM} is estimator of Dimson (1979), β^{BLU} is estimator of Blume (1975) (also known as Bloomberg β), β^{TOP} is β with respect to ten largest stocks by market capitalization, EW stands for expanding window, RW for rolling window. Changes in BMI and CAPM βs are winsorized at the 1% and 99% level to remove outliers. Standard errors in parentheses are clustered at firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A3: Placebo test using the CAPM β of peer firms

	(1) Firm's Δ CAPM β^E	(2) Firm's Δ CAPM β^E	(3) All Peers' Δ CAPM β^E	(4) Peer n=1's Δ CAPM β^E	(5) Peer n=2 Δ CAPM β^E	(6) Peer n=3's Δ CAPM β^E
Δ CAPM β^{Peer}	0.232***					
	(0.010)					
Δ BMI		0.814***	0.037	0.075	-0.009	0.043
		(0.054)	(0.029)	(0.046)	(0.043)	(0.045)
Fixed Effects						
Firm FE	✓	✓	✓	✓	✓	✓
Peer FE	✓		✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.421	0.249	0.472	0.466	0.471	0.479
Observations	46,689	16,995	47,749	16,470	16,022	15,257

Notes: This table reports coefficient estimates for a placebo test using N=3 firm peers' change in CAPM β and assigns them the Δ BMI of the firm: $\Delta\text{CAPM } \beta^{\text{Peer}}_{j,t} = \alpha_i + \alpha_j + \alpha_t + \Delta\text{BMI}^{\text{Firm}}_{i,t} + \varepsilon_{j,i,t}$ for firm i and peer j in year t. Standard errors in parentheses are clustered at the firm-level in column (2) and double clustered at firm and peer level in other columns. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A4: Changes in CAPM β and firm-level risk measures of Hassan et al. (2019)

(in σ units)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Δ Risk		Δ Pol. Risk		Δ Pol. Risk - Econ.		Δ Pol. Risk - Secu.		Δ Pol. Risk - Tech.		Δ Pol. Risk - Trade	
Δ CAPM β^E	0.0192**		0.0170*		0.0163*		0.0149*		0.0087		-0.0002	
	(0.007)		(0.007)		(0.007)		(0.007)		(0.007)		(0.007)	
Δ BMI		-0.0086		0.0080		-0.0011		0.0097		0.0115		-0.0088
		(0.009)		(0.009)		(0.009)		(0.009)		(0.009)		(0.009)
Fixed Effects												
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.14	0.14	0.13	0.13	0.13	0.13	0.13	0.13	0.12	0.12	0.13	0.13
Observations	29,970	29,970	29,985	29,985	29,963	29,963	29,978	29,978	29,982	29,982	29,976	29,976

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{Firm-level Risk}_{i,t} = \alpha_i + \alpha_t + \gamma\, \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Changes in firm-level risk (Hassan et al., 2019) calculated between 1st and 4th quarter of the year. Coefficients are standardized to unit variances. Changes in firm-level risk measures, CAPM βs, and BMI are trimmed at the 1% and 99% level to remove outliers. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A5: Changes in CDS Spreads and CAPM βs of CDS Spreads

Dependent variable:	Δ CDS Spread (in σ units)				Δ CDS CAPM β (in σ units)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ BMI (in σ units)	-0.0221	-0.0244	-0.0189	-0.0283	0.0280	0.0189	0.0014	0.0305
	(0.0212)	(0.0209)	(0.0207)	(0.0210)	(0.0240)	(0.0242)	(0.0253)	(0.0260)
Momentum (Cum. Ret.)		-0.120***	-0.140***			-0.0618**	-0.0366	
(in σ units)		(0.0315)	(0.0332)			(0.0238)	(0.0259)	
Fixed Effects								
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓			✓	✓		
Year × Size Decile			✓				✓	
Year × Momentum Decile				✓				✓
Adj. R^2	0.260	0.269	0.303	0.315	0.103	0.106	0.179	0.155
Observations	2,798	2,798	2,798	2,798	2,299	2,299	2,299	2,299

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{CDS Spreads}_{i,t} = \alpha_i + \alpha_t + \gamma\, \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Coefficients are standardized to unit variances. CDS spreads for senior unsecured debt with tenor of 5 year and doc clause XR14 (no restructuring). CDS CAPM βs are calculated on daily data from 2010 to 2019 using the weighted least squares estimator of Welch (2022b) with exponentially decay of 3 months half life. Changes in CDS spreads and CDS CAPM βs are trimmed at the 2% and 98% level to remove outliers. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A6: Changes in measures of text-based financial frictions (Hoberg and Maksimovic, 2015)

(in σ units)	(1) Δ Inv. Delay	(2) Δ Inv. Delay & Equity Issue	(3) Δ Inv. Delay & Debt Issue	(4) Δ Inv. Delay & Private Issue	(5) Δ Inv. Delay & Equity (LW, '23)	(6) Δ Inv. Delay & Debt (LW, '23)
Δ BMI	-0.0008	-0.007	-0.0004	-0.005	-0.010	0.004
	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.007)
Fixed Effects						
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.08	0.07	0.06	0.07	0.07	0.04
Observations	23,463	23,463	23,463	23,463	32,275	32,275

Notes: This table reports coefficients estimates for regression specifications of the form: ΔMeasure of Financial Constraint$_{i,t} = \alpha_i + \alpha_t + \gamma\, \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Changes in text-based financial constraint measures from Hoberg and Maksimovic (2015) and Linn and Weagley (2021). Coefficients are standardized to unit variances. Changes in financial constraints measures, CAPM βs, and BMI are trimmed at the 1% and 99% level to remove outliers. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A7: Changes in measures of corporate governance

(in σ units)	(1) Δ S&P G-Score	(2) Δ Sus. G-Score	(3) Δ Ref. G-Score	(4) Δ Sus. ESG	(5) Δ S&P ESG	(6) Δ Ref. ESG
Δ BMI	-0.024	-0.020	0.008	-0.009	0.031	-0.003
	(0.057)	(0.024)	(0.017)	(0.026)	(0.057)	(0.017)
Fixed Effects						
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.31	0.20	0.07	0.23	0.34	0.10
Observations	2,003	7,168	13,925	7,326	2,003	13,925

Notes: This table reports coefficients estimates for regression specifications of the form: ΔGovernance Score$_{i,t} = \alpha_i + \alpha_t + \gamma\, \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Governance and ESG scores of Standard & Poor, Sustainalytics, and Refinitiv. Coefficients are standardized to unit variances. Changes in G-Scores, ESG Scores, and BMI are trimmed at the 1% and 99% level to remove outliers. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A8: Lag-selection based on Bayesian/Schwartz Information Criterion

Maximum lag length	L=0	L=1	L=2	L=3	L=4	L=5	L=6
Bayesian Information Criterion (OLS)	451	391	389	310	187	**157**	165
Bayesian Information Criterion (IV)	1,970	755	518	460	**411**	577	628
Observations	14,495	14,495	14,495	14,495	14,495	14,495	14,495

Notes: This table reports BIC for regression specifications of the form: ΔPerceived Cost of Capital$_{i,t} = \alpha_i + \alpha_t + \gamma\ \Delta\text{CAPM}\ \beta_{i,t} + \nu_{i,t}$ for OLS and IV regression specification where changes in CAPM β are instrumented by changes in BMI.

A.3 RDD at the Russell 3000 cutoff

We employ an alternative strategy to estimate the change in a firm's CAPM β due to a change in the stock's BMI using a regression discontinuity design at the Russell 3000 cutoffs.[24] To do so, we translate monthly estimates of CAPM βs to the year-firm-level using the following procedure.

$$\Delta\text{CAPM } \beta = \text{Avg. post-period } \hat{\beta} - \text{Avg. pre-period } \hat{\beta}$$

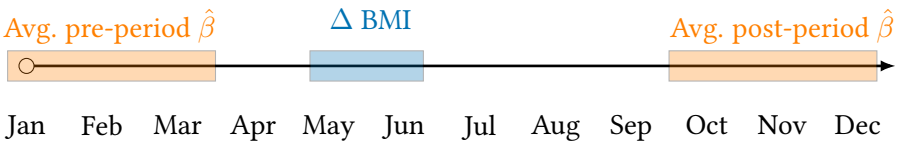


Figure A3: Timeline of Changes in CAPM

We define the change in a firm's CAPM β as the difference between the average β in the period before the BMI changes[25] (December to March) and a post-period (August to November). We drop April to July to avoid the potentially confounding effects of trading related to the index inclusion or exclusion on the β estimates. Figure A3 provides a visual representation of the timeline.

We estimate the local average treatment effect of Russell 3000 index inclusion on a firm's CAPM β by estimating a series of regression discontinuity specifications using the following form:

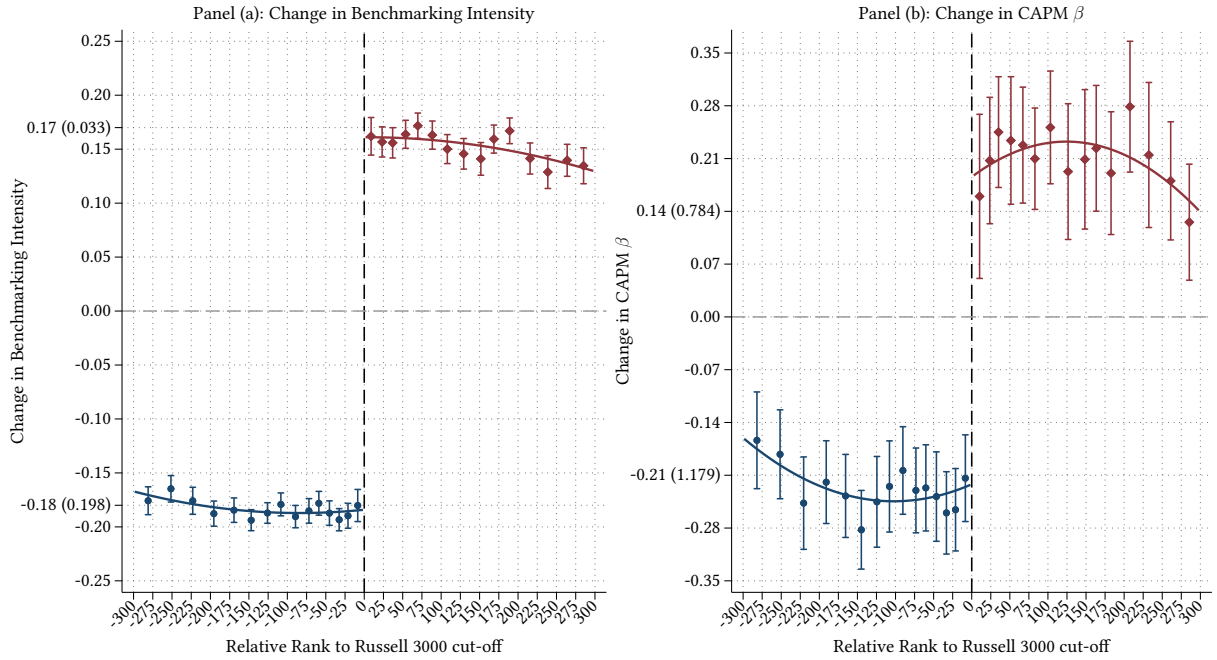
$$\Delta\text{CAPM } \beta_i = \alpha_0 + \alpha_1 D_i + f(\text{Rank} - \text{Cutoff})_i + f(\text{Rank} - \text{Cutoff})_i \times D_i + \eta_i \qquad (18)$$

where D_i is an indicator variable for whether firm i's market value rank exceeds the Russell 3000 inclusion cutoff in year t. The coefficient of interest, α_1, summarizes the local average treatment effect of changes in BMI due to index reconstitution on a firm's CAPM β.

Identifying assumptions and threats to identification Our identification assumes that firms are not able to manipulate their market equity rank relative to the index cutoff with precision. This assumption is typically tested by checking for bunching around the cutoff with the McCrary (2008) test. However, bunching is not possible because the assignment variable is a rank. Moreover, an idiosyncratic shock to the market value on the rank date can bring the stock to the other side of the cutoff. Taken together, these two points imply that index assignment near the cutoff

[24] Appendix A.4 repeats this exercise at the Russell 1000 cutoffs for the years from 2002 to 2006 with similar results.
[25] We again focus on stocks for which the BMI changes by at least 5 p.p. in absolute value to ensure that we select stocks that are affected by the index reconstitution.

Figure A4: Regression discontinuity study at Russell 3000 cutoffs



Notes: Sample from 2008 to 2018. Y-axis values in parentheses are average of variables in levels pre-treatment within 50 ranks of cutoff. Changes in BMI are calculated as the difference in BMI between May and June. Changes in CAPM β is difference between the average in the first and fourth quarter. Standard errors are clustered at firm-level.

is as good as random (Chang, Hong, and Liskovich, 2015). Limburg (2024) tests if firm try to manipulate their equity rank to become index members but finds no evidence of such behavior.

Results Panel (a) of Figure A4 shows that the BMI discontinuously changes at the Russell 3000 inclusion cutoff. The BMI for stocks included in the Russell 3000 increases on average by 17 p.p.; an approximately five-fold increase relative to their pre-treatment average (measured within 50 ranks of the cutoff) of 3% BMI. Conversely, stocks dropped from the Russell 3000 see an average decrease of 18 p.p. in BMI.

Panel (b) of Figure A4 shows that this increase in BMI associated with index reconstitution drastically change the CAPM βs of treated firms. The CAPM β of stocks that experience an increase in BMI due to index inclusion increases by 0.14 on average. Conversely, the CAPM β of stocks that experience a decrease in BMI at index reconstitution decreases by 0.21; a decrease of 18% relative to their pre-treatment average CAPM β of 1.18.

Table A9 reports the results of the regression discontinuity specification in (18). The coefficient estimate of the local average treatment effect, α_1, is positive and always statistically significant at the 5% level or higher. Columns (1) to (7) use a parametric regression discontinuity

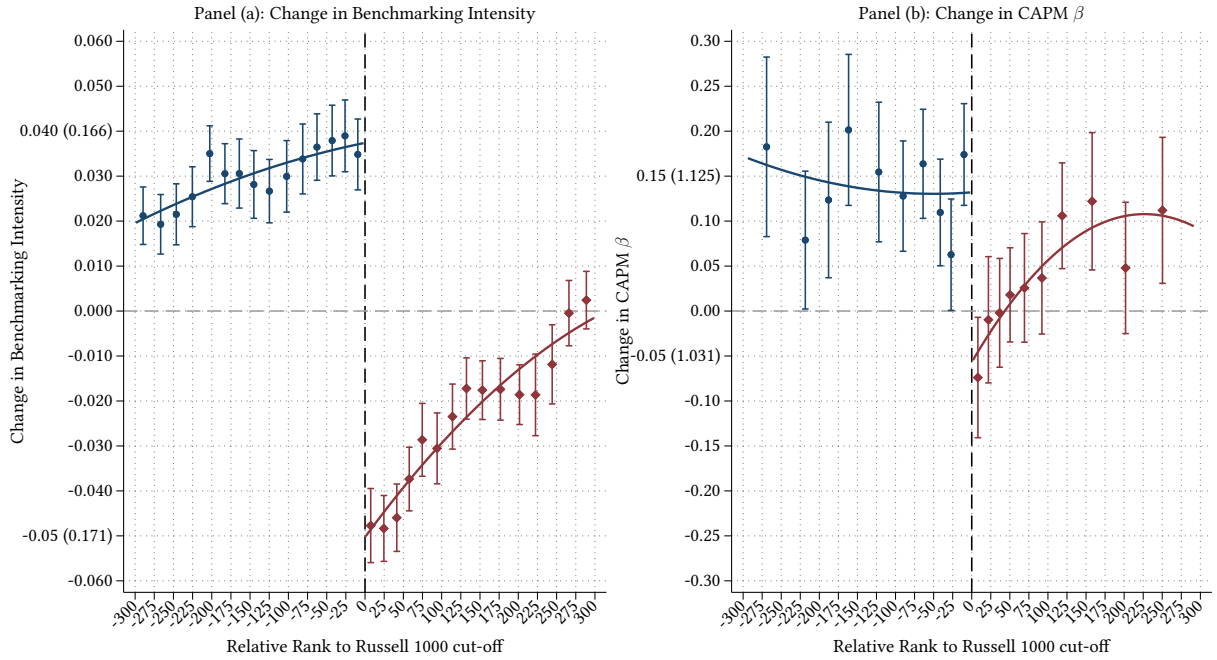
Table A9: Regression Discontinuity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: Δ CAPM β							
LATE	0.363***	0.305**	0.342**	0.311*	0.342**	0.289*	0.303***	0.218*
	(0.0740)	(0.0963)	(0.0880)	(0.105)	(0.0995)	(0.102)	(0.0783)	(0.101)
$f(X)$	Linear	Quadratic	Linear	Quadratic	Linear	Quadratic	Poly(1)	Poly(2)
Window width	300	300	200	200	150	150	300	300
Kernel							Triangular	Triangular
Bandwidth							122.4	111.2
Adj. R^2	0.415	0.417	0.440	0.441	0.437	0.440		
Observations	1,185	1,185	914	914	737	737	1,185	1,185

Notes: This table reports the local average treatment effect (α_1) of Russell 3000 index inclusion on a firm's CAPM β from regression discontinuity specifciations: $\Delta\text{CAPM } \beta_i = \alpha_0 + \alpha_1 D_i + f(\text{Rank} - \text{Cutoff})_i + f(\text{Rank} - \text{Cutoff})_i \times D_i + \eta_i$. Standard errors in parentheses are double clustered at firm and year level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

designs, while (7) and (8) are show non-parametric estimates using local polynomial regressions of order 1 and 2 using triangular kernels with automatic bandwidth selection. Columns (1), (2), (7), (8) estimate treatment effects around the index cutoffs with a window of 300 ranks on each side. Columns (3) and (4) use a window of 200 ranks, and columns (5) and (6) use a window of 150 ranks. The estimated coefficients are reasonably stable across columns (1) to (7) but around one third smaller for the local quadratic non-parametric estimate of column (8). The local average treatment effect, however, are again all large and economically meaningful. For example, the coefficient estimate of 0.289 in Column (6) implies an increase of 1.73 p.p. in the equity cost of capital using the CAPM (assuming an ERP of 6%).

Figure A5: Regression Discontinuity at Russell 1000 cutoffs



Notes: Sample from 2002 to 2006. Y-axis values in parentheses are averages of variables in levels pre-treatment within 50 ranks of cutoff.

A.4 RDD at the Russell 1000 cutoff

Table A10: Regression Discontinuity at Russell 1000 cutoffs

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: Δ CAPM β							
LATE	-0.114*	-0.142*	-0.124+	-0.131+	-0.126+	-0.167+	-0.233***	-0.262***
	(0.0354)	(0.0509)	(0.0551)	(0.0488)	(0.0490)	(0.0641)	(0.0573)	(0.0642)
f(X)	Linear	Quadratic	Linear	Quadratic	Linear	Quadratic	Poly(1)	Poly(2)
Window width	300	300	200	200	150	150	300	300
Kernel							Triangular	Triangular
Bandwidth							52.8	86.4
Adj. R^2	0.104	0.106	0.124	0.132	0.159	0.162		
Observations	474	474	390	390	322	322	474	474

Notes: This table reports the local average treatment effect (α_1) of Russell 1000 index inclusion on a firm's CAPM β from regression discontinuity specifciations: $\Delta\text{CAPM } \beta_i = \alpha_0 + \alpha_1 D_i + f(\text{Rank} - \text{Cutoff})_i + f(\text{Rank} - \text{Cutoff})_i \times D_i + \eta_i$. Standard errors in parentheses are double clustered at firm and year level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.